tropical: FloydWarshall

2026-10-24T16:14:17

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `765d817c3d73`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9074	[0.899, 0.914]	0.0097	5	0
adaptive	pointer	0.01	0.9093	[0.904, 0.914]	0.00625	5	0
adaptive	pointer	0.1	0.9091	[0.904, 0.914]	0.0063	5	0
adaptive	pointer	1	0.9092	[0.905, 0.913]	0.00574	5	0
adaptive	pointer	10	0.9088	[0.903, 0.914]	0.00692	5	0
adaptive	pointer	100	0.9093	[0.905, 0.914]	0.00609	5	0
adaptive	pointer	1000	0.9088	[0.904, 0.913]	0.00594	5	0
adaptive	pointer	1e+04	0.9081	[0.904, 0.912]	0.00533	5	0
adaptive	seq2seq	0	0.8813	[0.874, 0.889]	0.00973	5	0
adaptive	seq2seq	0.01	0.8753	[0.873, 0.878]	0.00333	5	0
adaptive	seq2seq	0.1	0.8762	[0.874, 0.879]	0.00379	5	0
adaptive	seq2seq	1	0.875	[0.873, 0.878]	0.00396	5	0
adaptive	seq2seq	10	0.8746	[0.873, 0.877]	0.00292	5	0
adaptive	seq2seq	100	0.8847	[0.873, 0.898]	0.0159	5	0
adaptive	seq2seq	1000	0.8866	[0.873, 0.901]	0.0185	5	0
adaptive	seq2seq	1e+04	0.8888	[0.858, 0.907]	0.0354	5	0
tropical	pointer	0	0.9135	[0.912, 0.915]	0.00176	5	0
tropical	pointer	0.01	0.9139	[0.913, 0.915]	0.00119	5	0
tropical	pointer	0.1	0.9124	[0.91, 0.914]	0.00241	5	0
tropical	pointer	1	0.9139	[0.913, 0.916]	0.00188	5	0
tropical	pointer	10	0.9134	[0.912, 0.915]	0.00162	5	0
tropical	pointer	100	0.914	[0.912, 0.916]	0.00197	5	0
tropical	pointer	1000	0.9139	[0.913, 0.915]	8.98×10^{-4}	5	0
tropical	pointer	1e+04	0.9134	[0.912, 0.915]	0.00195	5	0
tropical	seq2seq	0	0.8958	[0.888, 0.903]	0.01	5	0
tropical	seq2seq	0.01	0.9035	[0.897, 0.91]	0.00839	5	0
tropical	seq2seq	0.1	0.9032	[0.897, 0.907]	0.00699	5	0
tropical	seq2seq	1	0.901	[0.896, 0.906]	0.00612	5	0
tropical	seq2seq	10	0.9006	[0.894, 0.907]	0.00855	5	0
tropical	seq2seq	100	0.9006	[0.893, 0.908]	0.00993	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.9034	[0.898, 0.908]	0.00624	5	0
tropical	seq2seq	1e+04	0.8946	[0.887, 0.903]	0.0104	5	0
vanilla	pointer	0	0.9111	[0.907, 0.914]	0.00489	5	0
vanilla	pointer	0.01	0.9135	[0.911, 0.915]	0.00236	5	0
vanilla	pointer	0.1	0.9135	[0.911, 0.915]	0.00241	5	0
vanilla	pointer	1	0.9135	[0.911, 0.915]	0.00236	5	0
vanilla	pointer	10	0.9136	[0.912, 0.915]	0.00219	5	0
vanilla	pointer	100	0.9134	[0.911, 0.915]	0.00254	5	0
vanilla	pointer	1000	0.9134	[0.911, 0.915]	0.00271	5	0
vanilla	pointer	1e+04	0.9133	[0.911, 0.915]	0.00284	5	0
vanilla	seq2seq	0	0.8999	[0.885, 0.911]	0.0166	5	0
vanilla	seq2seq	0.01	0.8887	[0.871, 0.906]	0.0229	5	0
vanilla	seq2seq	0.1	0.8872	[0.87, 0.905]	0.0234	5	0
vanilla	seq2seq	1	0.8872	[0.869, 0.905]	0.0236	5	0
vanilla	seq2seq	10	0.8877	[0.87, 0.905]	0.0234	5	0
vanilla	seq2seq	100	0.8874	[0.869, 0.906]	0.0249	5	0
vanilla	seq2seq	1000	0.9029	[0.888, 0.912]	0.0168	5	0
vanilla	seq2seq	1e+04	0.9122	[0.91, 0.914]	0.00296	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.001874	[-0.00975, 0.0135]	0.9977
adaptive	pointer	0.1	0.001644	[-0.00998, 0.0133]	0.9990
adaptive	pointer	1	0.001812	[-0.00982, 0.0134]	0.9981
adaptive	pointer	10	0.001426	[-0.0102, 0.0131]	0.9996
adaptive	pointer	100	0.001921	[-0.00971, 0.0135]	0.9973
adaptive	pointer	1000	0.001405	[-0.0102, 0.013]	0.9996
adaptive	pointer	1e+04	6.698×10^{-4}	[-0.011, 0.0123]	1.0000
adaptive	seq2seq	0.01	-0.006069	[-0.0336, 0.0215]	0.9859
adaptive	seq2seq	0.1	-0.005151	[-0.0327, 0.0224]	0.9944
adaptive	seq2seq	1	-0.006328	[-0.0339, 0.0212]	0.9823
adaptive	seq2seq	10	-0.006693	[-0.0343, 0.0209]	0.9762
adaptive	seq2seq	100	0.003377	[-0.0242, 0.031]	0.9996
adaptive	seq2seq	1000	0.005214	[-0.0224, 0.0328]	0.9940
adaptive	seq2seq	1e+04	0.007491	[-0.0201, 0.0351]	0.9581
tropical	pointer	0.01	4.576×10^{-4}	[-0.00263, 0.00355]	0.9986
tropical	pointer	0.1	-0.001078	[-0.00417, 0.00201]	0.8738
tropical	pointer	1	4.429×10^{-4}	[-0.00265, 0.00353]	0.9989
tropical	pointer	10	-7.008×10^{-5}	[-0.00316, 0.00302]	1.0000
tropical	pointer	100	5.329×10^{-4}	[-0.00256, 0.00362]	0.9965
tropical	pointer	1000	4.158×10^{-4}	[-0.00267, 0.0035]	0.9992
tropical	pointer	1e+04	-8.520×10^{-5}	[-0.00317, 0.003]	1.0000
tropical	seq2seq	0.01	0.007759	[-0.00707, 0.0226]	0.5562
tropical	seq2seq	0.1	0.007434	[-0.0074, 0.0223]	0.5988
tropical	seq2seq	1	0.00526	[-0.00957, 0.0201]	0.8655
tropical	seq2seq	10	0.004858	[-0.00997, 0.0197]	0.9023
tropical	seq2seq	100	0.004797	[-0.01, 0.0196]	0.9073
tropical	seq2seq	1000	0.007602	[-0.00723, 0.0224]	0.5766
tropical	seq2seq	1e+04	-0.001179	[-0.016, 0.0137]	1.0000
vanilla	pointer	0.01	0.002386	[-0.00269, 0.00746]	0.6593
vanilla	pointer	0.1	0.002413	[-0.00266, 0.00749]	0.6490
vanilla	pointer	1	0.002382	[-0.00269, 0.00745]	0.6609

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	10	0.002494	[-0.00258, 0.00757]	0.6174
vanilla	pointer	100	0.002241	[-0.00283, 0.00731]	0.7147
vanilla	pointer	1000	0.002235	[-0.00284, 0.00731]	0.7170
vanilla	pointer	1e+04	0.002138	[-0.00293, 0.00721]	0.7533
vanilla	seq2seq	0.01	-0.01129	[-0.0471, 0.0245]	0.9172
vanilla	seq2seq	0.1	-0.0127	[-0.0485, 0.0231]	0.8657
vanilla	seq2seq	1	-0.0128	[-0.0486, 0.023]	0.8617
vanilla	seq2seq	10	-0.01225	[-0.0481, 0.0236]	0.8834
vanilla	seq2seq	100	-0.01256	[-0.0484, 0.0233]	0.8715
vanilla	seq2seq	1000	0.002902	[-0.0329, 0.0387]	1.0000
vanilla	seq2seq	1e+04	0.01227	[-0.0236, 0.0481]	0.8829

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-1.422×10^{-4}	$[-4.55 \times 10^{-4}, 6.48 \times 10^{-5}]$	0.7668	-0.059	0.6717
adaptive	seq2seq	0.00254	$[5.02 \times 10^{-4}, 0.00396]$	0.0601	0.147	0.2823
tropical	pointer	5.173×10^{-5}	$[-6.31 \times 10^{-5}, 1.74 \times 10^{-4}]$	0.7235	0.04	0.7688
tropical	seq2seq	-9.620×10^{-4}	$[-0.00173, -1.96 \times 10^{-4}]$	0.1598	-0.181	0.1776
vanilla	pointer	-4.433×10^{-5}	$[-1.24 \times 10^{-4}, 1.22 \times 10^{-5}]$	0.8210	-0.0476	0.7297
vanilla	seq2seq	0.003647	$[5.46 \times 10^{-4}, 0.00675]$	0.0408	0.173	0.1984

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.002332	[-0.00348, 0.00815]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	4.037×10^{-4}	[-0.00277, 0.00358]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.001159	[-0.00521, 0.00289]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	3.929×10^{-4}	[-0.00334, 0.00413]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-2.323×10^{-4}	[-0.00378, 0.00331]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	6.239×10^{-4}	[-0.00227, 0.00351]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	5.126×10^{-4}	[-0.00294, 0.00396]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	1.091×10^{-4}	[-0.00269, 0.00291]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.004183	[-0.032, 0.0236]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.01486	[-0.0174, 0.0472]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01595	[-0.0114, 0.0433]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01387	[-0.013, 0.0408]	1.0000

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01293	[-0.0204, 0.0462]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.01317	[-0.0212, 0.0475]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	5.174×10^{-4}	[-0.0256, 0.0266]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.01763	[-0.0285, -0.00673]	0.1746	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.003711	[-0.0159, 0.00845]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.004223	[-0.00994, 0.00149]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.00448	[-0.0105, 0.00153]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.004281	[-0.00971, 0.00115]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.004779	[-0.0114, 0.00181]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.004031	[-0.0101, 0.00201]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.004542	[-0.0101, 0.00106]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.005179	[-0.0102, -1.70×10^{-4}]	0.7270	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.01861	[-0.0403, 0.00308]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.01339	[-0.044, 0.0173]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01106	[-0.0417, 0.0196]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01214	[-0.0445, 0.0202]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.01305	[-0.0445, 0.0184]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.002676	[-0.0324, 0.0271]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.0163	[-0.055, 0.0224]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.02339	[-0.0691, 0.0223]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.02608	[0.00659, 0.0456]	0.3395	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.03402	[0.0263, 0.0417]	0.0058	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.03288	[0.0241, 0.0417]	0.0103	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.03422	[0.0272, 0.0412]	0.0041	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.0342	[0.0257, 0.0427]	0.0081	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.02463	[4.43×10^{-4}, 0.0488]	0.5561	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02227	[-0.00361, 0.0482]	0.7526	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01926	[-0.0211, 0.0596]	0.9511	

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0177	[0.00459, 0.0308]	0.3395
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.0104	$[5.42 \times 10^{-4}, 0.0203]$	0.5561
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.009186	$[4.84 \times 10^{-4}, 0.0179]$	0.5561
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01288	[0.00462, 0.0211]	0.2467
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.01277	[0.00207, 0.0235]	0.4427
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01343	[0.00198, 0.0249]	0.4427
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01051	[0.00312, 0.0179]	0.3035
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.01879	[0.0061, 0.0315]	0.2794
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01118	[-0.012, 0.0343]	0.9511
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.02486	[-0.00514, 0.0548]	0.7526
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0263	[-0.00434, 0.0569]	0.7526
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.02636	[-0.00447, 0.0572]	0.7526
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.02593	[-0.00447, 0.0563]	0.7526
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02598	[-0.00674, 0.0587]	0.7526
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01052	[-0.0105, 0.0316]	0.9511
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.001052	[-0.0052, 0.0073]	0.9511

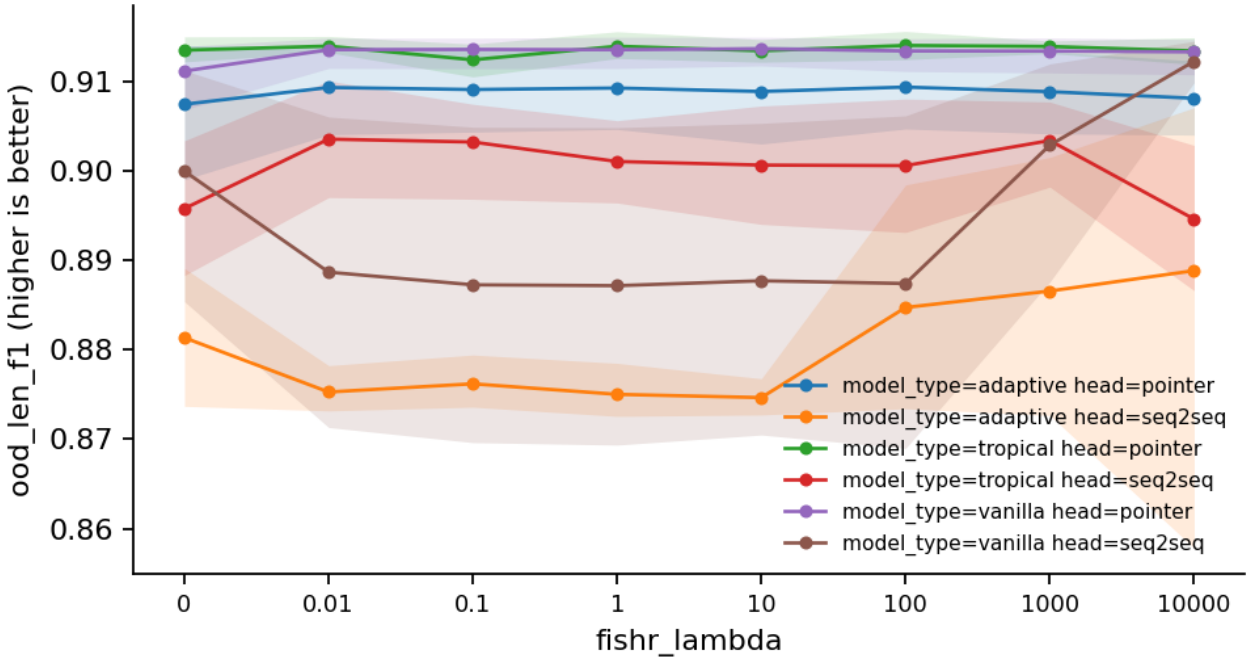


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

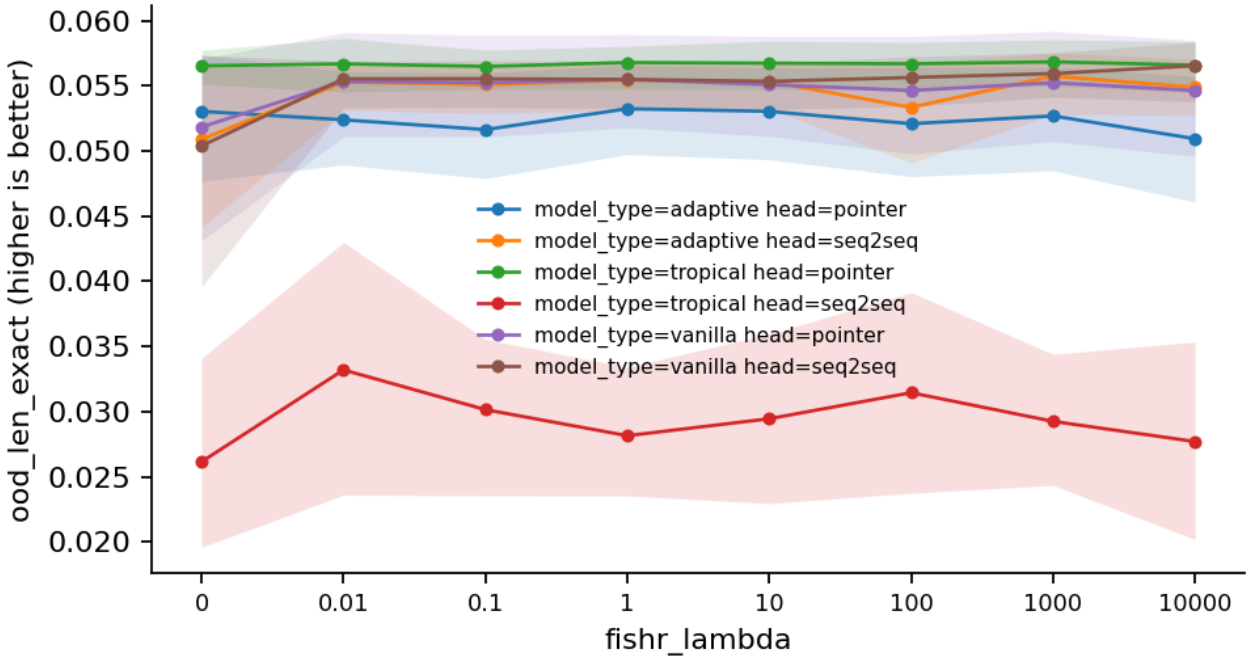


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.05305	[0.0477, 0.0574]	0.0065	5	0
adaptive	pointer	0.01	0.0524	[0.0489, 0.0561]	0.00479	5	0
adaptive	pointer	0.1	0.05165	[0.0479, 0.056]	0.00525	5	0
adaptive	pointer	1	0.05325	[0.0497, 0.0566]	0.00443	5	0
adaptive	pointer	10	0.05305	[0.0493, 0.0564]	0.00472	5	0
adaptive	pointer	100	0.0521	[0.048, 0.0565]	0.00548	5	0
adaptive	pointer	1000	0.0527	[0.0485, 0.0565]	0.00498	5	0
adaptive	pointer	1e+04	0.05095	[0.0461, 0.0557]	0.00634	5	0
adaptive	seq2seq	0	0.0509	[0.0442, 0.0566]	0.0079	5	0
adaptive	seq2seq	0.01	0.0553	[0.0531, 0.0568]	0.00241	5	0
adaptive	seq2seq	0.1	0.0551	[0.0529, 0.0566]	0.00245	5	0
adaptive	seq2seq	1	0.05545	[0.0531, 0.057]	0.00259	5	0
adaptive	seq2seq	10	0.05535	[0.0533, 0.0568]	0.00243	5	0
adaptive	seq2seq	100	0.05335	[0.0491, 0.0568]	0.00492	5	0
adaptive	seq2seq	1000	0.05575	[0.0529, 0.0576]	0.00315	5	0
adaptive	seq2seq	1e+04	0.0549	[0.0527, 0.0571]	0.00281	5	0
tropical	pointer	0	0.05655	[0.0551, 0.0578]	0.0018	5	0
tropical	pointer	0.01	0.0567	[0.0546, 0.0587]	0.00274	5	0
tropical	pointer	0.1	0.0565	[0.0548, 0.0578]	0.00197	5	0
tropical	pointer	1	0.0568	[0.0548, 0.0581]	0.00229	5	0
tropical	pointer	10	0.05675	[0.0548, 0.0585]	0.00237	5	0
tropical	pointer	100	0.0567	[0.0547, 0.0583]	0.00239	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.05685	[0.055, 0.0586]	0.0024	5	0
tropical	pointer	1e+04	0.0566	[0.0545, 0.0584]	0.0025	5	0
tropical	seq2seq	0	0.02615	[0.0196, 0.0341]	0.0091	5	0
tropical	seq2seq	0.01	0.0332	[0.0236, 0.043]	0.0129	5	0
tropical	seq2seq	0.1	0.03015	[0.0236, 0.0355]	0.00753	5	0
tropical	seq2seq	1	0.02815	[0.0235, 0.0334]	0.00654	5	0
tropical	seq2seq	10	0.02945	[0.0229, 0.036]	0.00834	5	0
tropical	seq2seq	100	0.03145	[0.0238, 0.0391]	0.0102	5	0
tropical	seq2seq	1000	0.02925	[0.0244, 0.0344]	0.0066	5	0
tropical	seq2seq	1e+04	0.0277	[0.0202, 0.0354]	0.0112	5	0
vanilla	pointer	0	0.05185	[0.0432, 0.0571]	0.00945	5	0
vanilla	pointer	0.01	0.05535	[0.0511, 0.0591]	0.00515	5	0
vanilla	pointer	0.1	0.05525	[0.0511, 0.0589]	0.00494	5	0
vanilla	pointer	1	0.0555	[0.0518, 0.0589]	0.00457	5	0
vanilla	pointer	10	0.0551	[0.0511, 0.0588]	0.00513	5	0
vanilla	pointer	100	0.05465	[0.0498, 0.0588]	0.00601	5	0
vanilla	pointer	1000	0.05525	[0.0507, 0.0592]	0.00543	5	0
vanilla	pointer	1e+04	0.05465	[0.0496, 0.0585]	0.00577	5	0
vanilla	seq2seq	0	0.0504	[0.0396, 0.0573]	0.0121	5	0
vanilla	seq2seq	0.01	0.05555	[0.0533, 0.0569]	0.00245	5	0
vanilla	seq2seq	0.1	0.05555	[0.0533, 0.057]	0.00244	5	0
vanilla	seq2seq	1	0.0555	[0.0533, 0.0569]	0.00241	5	0
vanilla	seq2seq	10	0.05535	[0.0532, 0.0568]	0.00235	5	0
vanilla	seq2seq	100	0.05565	[0.0534, 0.0573]	0.00258	5	0
vanilla	seq2seq	1000	0.05595	[0.0541, 0.0576]	0.00225	5	0
vanilla	seq2seq	1e+04	0.05655	[0.0538, 0.0584]	0.00308	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-6.500×10^{-4}	[-0.01, 0.00871]	1.0000
adaptive	pointer	0.1	-0.0014	[-0.0108, 0.00796]	0.9985
adaptive	pointer	1	2.000×10^{-4}	[-0.00916, 0.00956]	1.0000
adaptive	pointer	10	0	[-0.00936, 0.00936]	1.0000
adaptive	pointer	100	-9.500×10^{-4}	[-0.0103, 0.00841]	0.9999
adaptive	pointer	1000	-3.500×10^{-4}	[-0.00971, 0.00901]	1.0000
adaptive	pointer	1e+04	-0.0021	[-0.0115, 0.00726]	0.9843
adaptive	seq2seq	0.01	0.0044	[-0.00261, 0.0114]	0.3701
adaptive	seq2seq	0.1	0.0042	[-0.00281, 0.0112]	0.4170
adaptive	seq2seq	1	0.00455	[-0.00246, 0.0116]	0.3371
adaptive	seq2seq	10	0.00445	[-0.00256, 0.0115]	0.3589
adaptive	seq2seq	100	0.00245	[-0.00456, 0.00946]	0.8728
adaptive	seq2seq	1000	0.00485	[-0.00216, 0.0119]	0.2770
adaptive	seq2seq	1e+04	0.004	[-0.00301, 0.011]	0.4671
tropical	pointer	0.01	1.500×10^{-4}	[-0.00391, 0.00421]	1.0000
tropical	pointer	0.1	-5.000×10^{-5}	[-0.00411, 0.00401]	1.0000
tropical	pointer	1	2.500×10^{-4}	[-0.00381, 0.00431]	1.0000
tropical	pointer	10	2.000×10^{-4}	[-0.00386, 0.00426]	1.0000
tropical	pointer	100	1.500×10^{-4}	[-0.00391, 0.00421]	1.0000
tropical	pointer	1000	3.000×10^{-4}	[-0.00376, 0.00436]	1.0000
tropical	pointer	1e+04	5.000×10^{-5}	[-0.00401, 0.00411]	1.0000
tropical	seq2seq	0.01	0.00705	[-0.00918, 0.0233]	0.7291
tropical	seq2seq	0.1	0.004	[-0.0122, 0.0202]	0.9743

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	1	0.002	[-0.0142, 0.0182]	0.9996
tropical	seq2seq	10	0.0033	[-0.0129, 0.0195]	0.9909
tropical	seq2seq	100	0.0053	[-0.0109, 0.0215]	0.9037
tropical	seq2seq	1000	0.0031	[-0.0131, 0.0193]	0.9937
tropical	seq2seq	1e+04	0.00155	[-0.0147, 0.0178]	0.9999
vanilla	pointer	0.01	0.0035	[-0.00695, 0.0139]	0.8929
vanilla	pointer	0.1	0.0034	[-0.00705, 0.0138]	0.9051
vanilla	pointer	1	0.00365	[-0.0068, 0.0141]	0.8733
vanilla	pointer	10	0.00325	[-0.0072, 0.0137]	0.9218
vanilla	pointer	100	0.0028	[-0.00765, 0.0132]	0.9608
vanilla	pointer	1000	0.0034	[-0.00705, 0.0138]	0.9051
vanilla	pointer	1e+04	0.0028	[-0.00765, 0.0132]	0.9608
vanilla	seq2seq	0.01	0.00515	[-0.00341, 0.0137]	0.4130
vanilla	seq2seq	0.1	0.00515	[-0.00341, 0.0137]	0.4129
vanilla	seq2seq	1	0.0051	[-0.00346, 0.0137]	0.4229
vanilla	seq2seq	10	0.00495	[-0.00361, 0.0135]	0.4535
vanilla	seq2seq	100	0.00525	[-0.00331, 0.0138]	0.3934
vanilla	seq2seq	1000	0.00555	[-0.00301, 0.0141]	0.3383
vanilla	seq2seq	1e+04	0.00615	[-0.00241, 0.0147]	0.2439

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-1.214×10^{-4}	$[-3.84 \times 10^{-4}, 1.79 \times 10^{-5}]$	0.7675	-0.0248	0.8664
adaptive	seq2seq	-7.143×10^{-5}	$[-1.86 \times 10^{-4}, 2.14 \times 10^{-5}]$	0.7749	0.0819	0.5397
tropical	pointer	1.071×10^{-5}	$[-3.57 \times 10^{-5}, 6.07 \times 10^{-5}]$	0.9545	0.0152	0.9150
tropical	seq2seq	-5.357×10^{-4}	$[-0.00169, 6.00 \times 10^{-4}]$	0.4718	-0.0762	0.5745
vanilla	pointer	-1.054×10^{-4}	$[-3.00 \times 10^{-4}, 4.82 \times 10^{-5}]$	0.8005	-0.0248	0.8580
vanilla	seq2seq	1.411×10^{-4}	$[6.61 \times 10^{-5}, 2.11 \times 10^{-4}]$	0.4798	0.137	0.2939

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0047	[-0.00821, 0.0176]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.00135	[-0.00345, 0.00615]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.00125	[-0.00377, 0.00627]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.0013	[-0.00369, 0.00629]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.00165	[-0.00421, 0.00751]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.00205	[-0.00435, 0.00845]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0016	[-0.00375, 0.00695]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.00195	[-0.00403, 0.00793]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.02425	[-0.0405, -0.00797]	0.1442	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.02235	[-0.0398, -0.00492]	0.2122	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.0254	[-0.0367, -0.0141]	0.0419	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.02735	[-0.0372, -0.0175]	0.0226	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.0259	[-0.0364, -0.0154]	0.0333	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.0242	[-0.0374, -0.011]	0.0776	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0267	[-0.0358, -0.0176]	0.0200	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.02885	[-0.0415, -0.0162]	0.0419	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0012	[-0.00329, 0.00569]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.00295	[-0.0127, 0.00678]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.0036	[-0.0136, 0.0064]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.00225	[-0.0114, 0.00686]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.00205	[-0.0122, 0.00807]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.00255	[-0.0144, 0.00932]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.00255	[-0.0129, 0.0078]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.0037	[-0.0152, 0.00781]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	5.000×10^{-4}	[-0.019, 0.02]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-2.500×10^{-4}	$[-9.78 \times 10^{-4}, 4.78 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-4.500×10^{-4}	$[-0.00105, 1.47 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-5.000×10^{-5}	$[-6.47 \times 10^{-4}, 5.47 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0	$[-5.81 \times 10^{-4}, 5.81 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0023	[-0.00805, 0.00345]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-2.000×10^{-4}	[-0.00249, 0.00209]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00165	[-0.0049, 0.0016]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.00215	[-0.0116, 0.0159]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.0029	[-0.00982, 0.00402]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.00345	[-0.0108, 0.00389]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.0022	[-0.0093, 0.0049]	1.0000	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0023	[-0.0098, 0.0052]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.00125	[-0.0129, 0.0104]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.00305	[-0.0107, 0.00462]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.00395	[-0.0119, 0.00399]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0304	[0.0192, 0.0416]	0.0366	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.0235	[0.00705, 0.04]	0.2821	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.02635	[0.0156, 0.0371]	0.0480	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.02865	[0.0188, 0.0385]	0.0293	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.0273	[0.0166, 0.038]	0.0435	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.02525	[0.0112, 0.0393]	0.1374	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.0276	[0.02, 0.0352]	0.0133	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.0289	[0.0147, 0.0431]	0.0911	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.00145	[-0.0209, 0.0238]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-2.000×10^{-4}	[-0.0063, 0.0059]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-3.000×10^{-4}	[-0.00596, 0.00536]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-1.388×10^{-18}	[-0.00534, 0.00534]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-2.500×10^{-4}	[-0.00621, 0.00571]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.001	[-0.00919, 0.00719]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-7.000×10^{-4}	[-0.00631, 0.00491]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.0019	[-0.00967, 0.00587]	1.0000	

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9985	[0.996, 1]	0.00289	5	0
adaptive	pointer	0.01	0.9975	[0.995, 1]	0.00294	5	0
adaptive	pointer	0.1	0.9977	[0.996, 1]	0.00243	5	0
adaptive	pointer	1	0.9985	[0.997, 1]	0.00204	5	0
adaptive	pointer	10	0.9979	[0.996, 1]	0.00259	5	0
adaptive	pointer	100	0.9985	[0.997, 1]	0.00192	5	0
adaptive	pointer	1000	0.9981	[0.996, 0.999]	0.00199	5	0
adaptive	pointer	1e+04	0.9987	[0.997, 1]	0.00169	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	0.1969	[0.0961, 0.301]	0.133	5	0
adaptive	seq2seq	0.01	0.3504	[0.174, 0.575]	0.258	5	0
adaptive	seq2seq	0.1	0.3871	[0.193, 0.615]	0.255	5	0
adaptive	seq2seq	1	0.3101	[0.177, 0.488]	0.211	5	0
adaptive	seq2seq	10	0.3469	[0.204, 0.552]	0.233	5	0
adaptive	seq2seq	100	0.2118	[0.102, 0.33]	0.151	5	0
adaptive	seq2seq	1000	0.2001	[0.113, 0.29]	0.114	5	0
adaptive	seq2seq	1e+04	0.2321	[0.151, 0.325]	0.114	5	0
tropical	pointer	0	0.9702	[0.945, 0.996]	0.033	5	0
tropical	pointer	0.01	0.999	[0.998, 1]	0.00104	5	0
tropical	pointer	0.1	0.9965	[0.993, 0.999]	0.00425	5	0
tropical	pointer	1	0.8704	[0.662, 0.985]	0.23	5	0
tropical	pointer	10	0.9867	[0.974, 0.996]	0.0143	5	0
tropical	pointer	100	0.804	[0.518, 0.993]	0.311	5	0
tropical	pointer	1000	0.9565	[0.89, 1]	0.0785	5	0
tropical	pointer	1e+04	0.9204	[0.765, 0.999]	0.174	5	0
tropical	seq2seq	0	0.5362	[0.198, 0.874]	0.445	5	0
tropical	seq2seq	0.01	0.8461	[0.719, 0.963]	0.167	5	0
tropical	seq2seq	0.1	0.633	[0.272, 0.975]	0.455	5	0
tropical	seq2seq	1	0.7754	[0.477, 1]	0.334	5	0
tropical	seq2seq	10	0.7709	[0.596, 0.955]	0.23	5	0
tropical	seq2seq	100	0.4134	[0.0748, 0.758]	0.44	5	0
tropical	seq2seq	1000	0.9332	[0.829, 0.997]	0.115	5	0
tropical	seq2seq	1e+04	0.9163	[0.764, 1]	0.167	5	0
vanilla	pointer	0	0.9909	[0.985, 0.997]	0.00773	5	0
vanilla	pointer	0.01	0.9956	[0.991, 0.999]	0.0052	5	0
vanilla	pointer	0.1	0.9922	[0.984, 1]	0.0108	5	0
vanilla	pointer	1	0.99	[0.979, 0.999]	0.013	5	0
vanilla	pointer	10	0.9882	[0.974, 0.999]	0.016	5	0
vanilla	pointer	100	0.9908	[0.982, 1]	0.0118	5	0
vanilla	pointer	1000	0.9894	[0.98, 0.998]	0.0119	5	0
vanilla	pointer	1e+04	0.995	[0.99, 0.999]	0.00584	5	0
vanilla	seq2seq	0	0.2159	[0.147, 0.306]	0.105	5	0
vanilla	seq2seq	0.01	0.3873	[0.217, 0.542]	0.209	5	0
vanilla	seq2seq	0.1	0.3444	[0.181, 0.508]	0.21	5	0
vanilla	seq2seq	1	0.3126	[0.166, 0.459]	0.191	5	0
vanilla	seq2seq	10	0.3357	[0.212, 0.474]	0.17	5	0
vanilla	seq2seq	100	0.4021	[0.234, 0.533]	0.191	5	0
vanilla	seq2seq	1000	0.2817	[0.174, 0.403]	0.147	5	0
vanilla	seq2seq	1e+04	0.1753	[0.103, 0.258]	0.0989	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-5.508×10^{-4}	[-0.00218, 2.50×10^{-5}]	1.0000
adaptive	pointer	0.1	-2.874×10^{-4}	[-0.00316, 0.00128]	1.0000
adaptive	pointer	1	-9.610×10^{-5}	[-9.38×10^{-4}, 0.00101]	1.0000
adaptive	pointer	10	-9.270×10^{-5}	[-0.00165, 3.30×10^{-4}]	1.0000
adaptive	pointer	100	-1.298×10^{-4}	[-9.13×10^{-4}, 9.72×10^{-4}]	1.0000
adaptive	pointer	1000	-3.555×10^{-4}	[-0.00143, 7.50×10^{-4}]	1.0000
adaptive	pointer	1e+04	3.090×10^{-6}	[-0.0019, 0.00222]	1.0000
adaptive	seq2seq	0.01	0.1463	[-0.0287, 0.342]	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	seq2seq	0.1	0.1325	[-0.0157, 0.391]	1.0000
adaptive	seq2seq	1	0.1006	[-0.0173, 0.25]	1.0000
adaptive	seq2seq	10	0.1654	$[-2.31 \times 10^{-4}$, 0.304]	1.0000
adaptive	seq2seq	100	-0.05391	[-0.154, 0.218]	1.0000
adaptive	seq2seq	1000	-1.478×10^{-4}	[-0.11, 0.151]	1.0000
adaptive	seq2seq	1e+04	-0.0307	[-0.129, 0.202]	1.0000
tropical	pointer	0.01	0.01565	[0.00311, 0.0544]	1.0000
tropical	pointer	0.1	0.0154	[0.00366, 0.0489]	0.8750
tropical	pointer	1	-0.03139	[-0.299, 0.0236]	1.0000
tropical	pointer	10	0.01558	[-0.00208, 0.0368]	1.0000
tropical	pointer	100	-0.006786	[-0.429, 0.0214]	1.0000
tropical	pointer	1000	0.0157	[-0.101, 0.0545]	1.0000
tropical	pointer	1e+04	0.01528	[-0.221, 0.0543]	1.0000
tropical	seq2seq	0.01	0.2234	[0.0447, 0.591]	1.0000
tropical	seq2seq	0.1	0	[-0.158, 0.403]	1.0000
tropical	seq2seq	1	0	[-0.0465, 0.638]	1.0000
tropical	seq2seq	10	0.2554	[-0.185, 0.678]	1.0000
tropical	seq2seq	100	0	[-0.572, 0.218]	1.0000
tropical	seq2seq	1000	0.3605	[0.072, 0.729]	1.0000
tropical	seq2seq	1e+04	0.2485	[0.0495, 0.729]	1.0000
vanilla	pointer	0.01	0.004756	[-0.0037, 0.0128]	1.0000
vanilla	pointer	0.1	1.215×10^{-4}	[-0.0096, 0.0121]	1.0000
vanilla	pointer	1	1.588×10^{-4}	[-0.0126, 0.0116]	1.0000
vanilla	pointer	10	-5.296×10^{-5}	[-0.0199, 0.0108]	1.0000
vanilla	pointer	100	4.984×10^{-5}	[-0.0123, 0.0119]	1.0000
vanilla	pointer	1000	2.336×10^{-4}	[-0.0128, 0.0104]	1.0000
vanilla	pointer	1e+04	1.464×10^{-4}	[-0.00351, 0.0117]	1.0000
vanilla	seq2seq	0.01	0.1753	[0.0564, 0.289]	0.4375
vanilla	seq2seq	0.1	0.05913	[0.0214, 0.277]	0.5000
vanilla	seq2seq	1	0.005633	[0.00382, 0.201]	0.4375
vanilla	seq2seq	10	0.1114	[0.0491, 0.19]	0.4375
vanilla	seq2seq	100	0.1501	[0.0778, 0.294]	0.5000
vanilla	seq2seq	1000	0.02088	[0.00523, 0.144]	0.5000
vanilla	seq2seq	1e+04	-0.03068	[-0.166, 0.0607]	1.0000

Table 11: `ood_val_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.1167	[-0.151, 0.382]	0.5044	0.0838	0.5371
adaptive	seq2seq	-0.2645	[-0.682, 0.232]	0.1247	-0.2	0.1416
tropical	pointer	-0.1584	[-0.567, 0.144]	0.3634	-0.131	0.3311
tropical	seq2seq	0.1176	[-0.154, 0.364]	0.5010	0.0857	0.5149
vanilla	pointer	-0.04526	[-0.22, 0.132]	0.7963	-0.0438	0.7552
vanilla	seq2seq	-0.2942	[-0.607, -0.0085]	0.0862	-0.234	0.0792

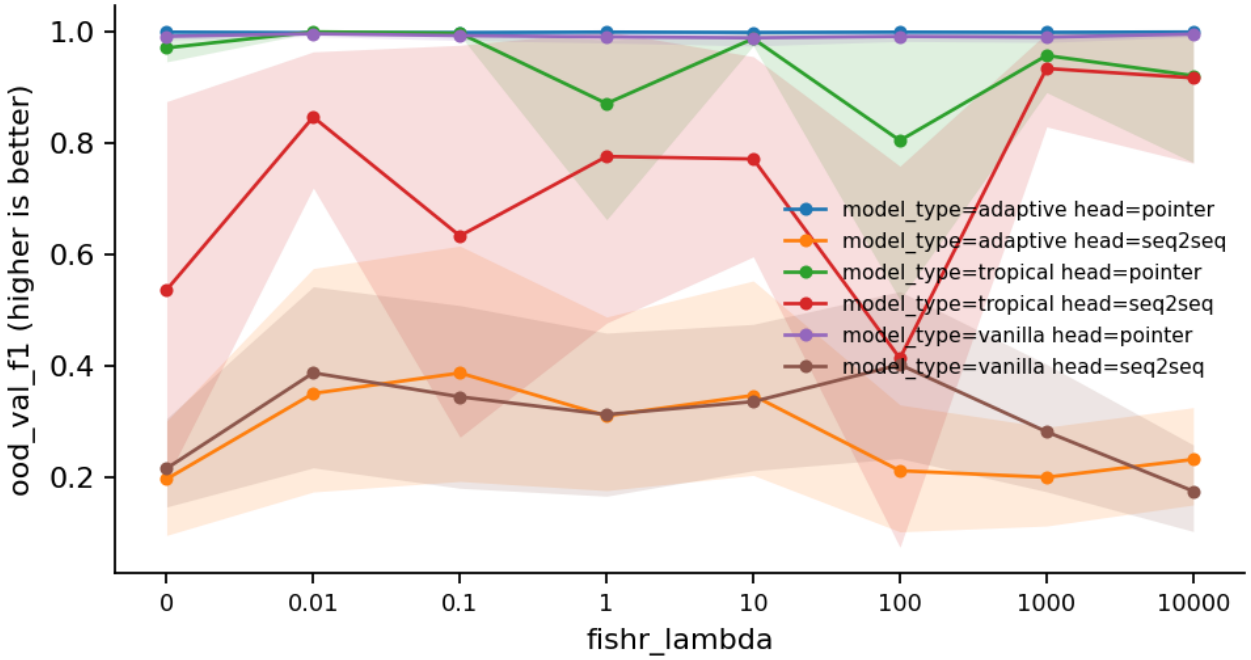


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.02073	[-0.0578, 0.0163]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.003361	[-0.00377, 0.0105]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.004272	[-0.00687, 0.0154]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.1196	[-0.412, 0.173]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.001476	[-0.0295, 0.0265]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.1868	[-0.581, 0.207]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.0329	[-0.124, 0.0582]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.07459	[-0.287, 0.138]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.3202	[-0.153, 0.793]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.4588	[0.118, 0.799]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.2886	[-0.159, 0.736]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.4628	[-0.0977, 1.02]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.4352	[0.113, 0.758]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.01129	[-0.378, 0.4]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.6515	[0.385, 0.919]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.7411	[0.519, 0.963]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.007566	[-0.00459, 0.0197]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.001857	[-0.00114, 0.00485]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.005543	[-0.0049, 0.016]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00854	[-0.00694, 0.024]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.009736	[-0.00735, 0.0268]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.007635	[-0.00538, 0.0207]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.008737	[-0.00414, 0.0216]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.003706	[-0.00367, 0.0111]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.01903	[-0.19, 0.152]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.03689	[-0.35, 0.276]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.04273	[-0.304, 0.389]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.002543	[-0.185, 0.179]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01123	[-0.178, 0.2]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.1902	[-0.473, 0.0927]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.08162	[-0.31, 0.146]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0568	[-0.152, 0.266]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.8016	[0.635, 0.968]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.647	[0.324, 0.97]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.6106	[0.293, 0.928]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.6885	[0.425, 0.952]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.651	[0.36, 0.943]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.7866	[0.601, 0.972]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.7981	[0.658, 0.938]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.7666	[0.625, 0.908]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.434	[-0.119, 0.987]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.1528	[-0.0549, 0.361]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.3635	[-0.202, 0.929]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.095	[-0.493, 0.683]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.2158	[-0.0791, 0.511]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.3906	[-0.138, 0.919]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.02327	[-0.179, 0.226]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.00403	[-0.335, 0.343]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.775	[0.65, 0.9]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.6083	[0.345, 0.871]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.6478	[0.389, 0.907]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.6774	[0.438, 0.917]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.6525	[0.426, 0.879]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.5887	[0.349, 0.828]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.7077	[0.522, 0.894]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.8197	[0.697, 0.943]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9939	[0.984, 1]	0.011	5	0
adaptive	pointer	0.01	0.9886	[0.978, 0.998]	0.0128	5	0
adaptive	pointer	0.1	0.9879	[0.972, 0.998]	0.0172	5	0
adaptive	pointer	1	0.9957	[0.993, 0.998]	0.00362	5	0
adaptive	pointer	10	0.9921	[0.987, 0.998]	0.00713	5	0
adaptive	pointer	100	0.9933	[0.988, 0.999]	0.0069	5	0
adaptive	pointer	1000	0.9915	[0.985, 0.997]	0.00795	5	0
adaptive	pointer	1e+04	0.9945	[0.99, 1]	0.00667	5	0
adaptive	seq2seq	0	0.04928	[0.0283, 0.0703]	0.0268	5	0
adaptive	seq2seq	0.01	0.1415	[0.047, 0.31]	0.187	5	0
adaptive	seq2seq	0.1	0.1642	[0.0616, 0.314]	0.168	5	0
adaptive	seq2seq	1	0.1294	[0.0511, 0.271]	0.157	5	0
adaptive	seq2seq	10	0.1221	[0.0435, 0.256]	0.149	5	0
adaptive	seq2seq	100	0.04788	[0.0209, 0.0798]	0.0376	5	0
adaptive	seq2seq	1000	0.02922	[0.016, 0.0448]	0.0188	5	0
adaptive	seq2seq	1e+04	0.02212	[0.0161, 0.0307]	0.00996	5	0
tropical	pointer	0	0.8602	[0.723, 0.972]	0.161	5	0
tropical	pointer	0.01	0.9929	[0.988, 0.998]	0.00701	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.9715	[0.942, 0.997]	0.0345	5	0
tropical	pointer	1	0.6795	[0.479, 0.859]	0.256	5	0
tropical	pointer	10	0.8913	[0.81, 0.972]	0.104	5	0
tropical	pointer	100	0.6042	[0.243, 0.939]	0.458	5	0
tropical	pointer	1000	0.8212	[0.558, 0.997]	0.312	5	0
tropical	pointer	1e+04	0.8138	[0.46, 0.996]	0.395	5	0
tropical	seq2seq	0	0.4825	[0.123, 0.842]	0.478	5	0
tropical	seq2seq	0.01	0.5548	[0.241, 0.853]	0.402	5	0
tropical	seq2seq	0.1	0.4912	[0.141, 0.844]	0.471	5	0
tropical	seq2seq	1	0.6379	[0.274, 1]	0.496	5	0
tropical	seq2seq	10	0.5242	[0.17, 0.885]	0.455	5	0
tropical	seq2seq	100	0.2472	[0.0275, 0.635]	0.424	5	0
tropical	seq2seq	1000	0.7152	[0.423, 0.952]	0.348	5	0
tropical	seq2seq	1e+04	0.8206	[0.587, 1]	0.264	5	0
vanilla	pointer	0	0.9556	[0.926, 0.987]	0.0396	5	0
vanilla	pointer	0.01	0.9825	[0.968, 0.997]	0.0189	5	0
vanilla	pointer	0.1	0.9574	[0.911, 0.999]	0.0638	5	0
vanilla	pointer	1	0.9464	[0.886, 0.997]	0.0801	5	0
vanilla	pointer	10	0.9559	[0.918, 0.993]	0.0494	5	0
vanilla	pointer	100	0.9526	[0.893, 0.998]	0.067	5	0
vanilla	pointer	1000	0.9447	[0.883, 0.993]	0.0709	5	0
vanilla	pointer	1e+04	0.9718	[0.938, 0.996]	0.0383	5	0
vanilla	seq2seq	0	0.08902	[0.0171, 0.186]	0.108	5	0
vanilla	seq2seq	0.01	0.2354	[0.0586, 0.406]	0.222	5	0
vanilla	seq2seq	0.1	0.2246	[0.0532, 0.385]	0.216	5	0
vanilla	seq2seq	1	0.2017	[0.0431, 0.36]	0.202	5	0
vanilla	seq2seq	10	0.2003	[0.0569, 0.346]	0.187	5	0
vanilla	seq2seq	100	0.2541	[0.0952, 0.413]	0.212	5	0
vanilla	seq2seq	1000	0.09448	[0.0175, 0.24]	0.162	5	0
vanilla	seq2seq	1e+04	0.01832	[0.0145, 0.0222]	0.00499	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.0011	[-0.0175, 0.00418]	1.0000
adaptive	pointer	0.1	-0.0023	[-0.0243, 0.00756]	1.0000
adaptive	pointer	1	-7.000×10^{-4}	[-0.00418, 0.0092]	1.0000
adaptive	pointer	10	-4.000×10^{-4}	[-0.00832, 0.0048]	1.0000
adaptive	pointer	100	-6.000×10^{-4}	[-0.00724, 0.00602]	1.0000
adaptive	pointer	1000	-0.0029	[-0.011, 0.00586]	1.0000
adaptive	pointer	1e+04	0	[-0.00688, 0.00766]	1.0000
adaptive	seq2seq	0.01	0.0328	[-0.00556, 0.249]	1.0000
adaptive	seq2seq	0.1	0.0496	[0.00608, 0.247]	1.0000
adaptive	seq2seq	1	0.0332	$[3.78 \times 10^{-4}$, 0.208]	1.0000
adaptive	seq2seq	10	0.0211	[-0.00341, 0.192]	1.0000
adaptive	seq2seq	100	0.0019	[-0.0404, 0.0409]	1.0000
adaptive	seq2seq	1000	-0.0132	[-0.0437, -0.00482]	0.4375
adaptive	seq2seq	1e+04	-0.0282	[-0.0527, 3.60×10^{-4}]	1.0000
tropical	pointer	0.01	0.1048	[0.021, 0.267]	0.8750
tropical	pointer	0.1	0.1035	[0.0238, 0.227]	0.8750
tropical	pointer	1	-0.1558	[-0.375, 0.0209]	1.0000
tropical	pointer	10	0.0717	[-0.0761, 0.119]	1.0000

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
tropical	pointer	100	-0.0836	[-0.548, 0.0179]	1.0000
tropical	pointer	1000	0.1049	[-0.386, 0.245]	1.0000
tropical	pointer	1e+04	0.1038	[-0.482, 0.258]	1.0000
tropical	seq2seq	0.01	0	[-0.0476, 0.245]	1.0000
tropical	seq2seq	0.1	0	[-0.429, 0.481]	1.0000
tropical	seq2seq	1	0	[-0.0819, 0.57]	1.0000
tropical	seq2seq	10	0	[-0.488, 0.589]	1.0000
tropical	seq2seq	100	-0.058	[-0.611, -0.00876]	1.0000
tropical	seq2seq	1000	0	[-0.12, 0.585]	1.0000
tropical	seq2seq	1e+04	0.2064	[-0.0541, 0.76]	1.0000
vanilla	pointer	0.01	0.0348	[-0.00312, 0.0569]	1.0000
vanilla	pointer	0.1	0.001	[-0.0413, 0.0467]	1.0000
vanilla	pointer	1	0.001	[-0.0625, 0.0445]	1.0000
vanilla	pointer	10	9.000×10^{-4}	[-0.0435, 0.039]	1.0000
vanilla	pointer	100	9.000×10^{-4}	[-0.0482, 0.046]	1.0000
vanilla	pointer	1000	0.0013	[-0.0575, 0.0373]	1.0000
vanilla	pointer	1e+04	0.0013	[-0.00774, 0.0465]	1.0000
vanilla	seq2seq	0.01	0.1779	[-0.00382, 0.303]	1.0000
vanilla	seq2seq	0.1	0.157	[-0.0132, 0.309]	1.0000
vanilla	seq2seq	1	0.1231	[-0.0201, 0.262]	1.0000
vanilla	seq2seq	10	0.1663	[-0.0202, 0.231]	1.0000
vanilla	seq2seq	100	0.1419	[-0.0118, 0.342]	1.0000
vanilla	seq2seq	1000	2.000×10^{-4}	[-0.0637, 0.0737]	1.0000
vanilla	seq2seq	1e+04	-0.0026	[-0.15, 0.0018]	1.0000

Table 15: `ood_val_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.1054	[-0.0851, 0.343]	0.5467	-0.0724	0.6001
adaptive	seq2seq	-0.6344	[-0.802, -0.407]	$< 10^{-4}$	-0.503	0.0004
tropical	pointer	-0.1782	[-0.567, 0.116]	0.3057	-0.15	0.2629
tropical	seq2seq	0.1162	[-0.106, 0.321]	0.5064	0.0705	0.5871
vanilla	pointer	-0.0573	[-0.292, 0.207]	0.7437	-0.0533	0.6985
vanilla	seq2seq	-0.4025	[-0.719, -0.197]	0.0165	-0.32	0.0170

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.09536	[-0.294, 0.104]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01038	[-0.0168, 0.0375]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01412	[-0.0501, 0.0784]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.2669	[-0.645, 0.112]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.06462	[-0.189, 0.0596]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.3484	[-0.972, 0.276]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.1235	[-0.512, 0.265]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.158	[-0.65, 0.334]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.3935	[-0.0833, 0.87]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.3194	[-0.267, 0.905]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.2666	[-0.108, 0.641]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.4362	[-0.303, 1.18]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.3239	[-0.224, 0.872]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.00682	[-0.473, 0.46]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.6207	[0.228, 1.01]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.8022	[0.475, 1.13]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.03828	[-0.0197, 0.0962]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00608	[-0.00354, 0.0157]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.03052	[-0.0279, 0.089]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.04928	[-0.0507, 0.149]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.03618	[-0.0195, 0.0918]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.04068	[-0.035, 0.116]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.04676	[-0.0317, 0.125]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02272	[-0.0263, 0.0717]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.03974	[-0.147, 0.0673]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.09382	[-0.302, 0.114]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.06042	[-0.328, 0.207]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.07236	[-0.271, 0.126]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.07826	[-0.221, 0.0641]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.2062	[-0.448, 0.0357]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.06526	[-0.279, 0.148]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0038	[-0.0116, 0.0192]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.9446	[0.901, 0.988]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.847	[0.612, 1.08]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.8237	[0.618, 1.03]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.8663	[0.668, 1.06]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.8701	[0.68, 1.06]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.9454	[0.904, 0.987]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.9623	[0.933, 0.991]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.9724	[0.955, 0.99]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.3778	[-0.166, 0.922]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.4381	[-0.0587, 0.935]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.4803	[-0.0805, 1.04]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.04158	[-0.75, 0.833]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.3671	[-0.304, 1.04]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.3569	[-0.278, 0.992]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.106	[-0.622, 0.834]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.00672	[-0.694, 0.681]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.8666	[0.732, 1]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.7471	[0.471, 1.02]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.7328	[0.48, 0.985]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.7447	[0.504, 0.985]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.7556	[0.51, 1]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.6985	[0.456, 0.942]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.8502	[0.616, 1.08]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.9535	[0.904, 1]	1.0000

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6138	[0.609, 0.618]	0.00555	5	0
adaptive	pointer	0.01	0.6108	[0.607, 0.615]	0.0051	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0.1	0.6114	[0.608, 0.616]	0.00525	5	0
adaptive	pointer	1	0.6092	[0.605, 0.614]	0.00588	5	0
adaptive	pointer	10	0.6106	[0.607, 0.615]	0.00553	5	0
adaptive	pointer	100	0.6145	[0.611, 0.619]	0.00534	5	0
adaptive	pointer	1000	0.6116	[0.609, 0.616]	0.0046	5	0
adaptive	pointer	1e+04	0.6147	[0.61, 0.619]	0.00597	5	0
adaptive	seq2seq	0	0.6231	[0.609, 0.637]	0.0168	5	0
adaptive	seq2seq	0.01	0.6238	[0.618, 0.631]	0.00813	5	0
adaptive	seq2seq	0.1	0.6244	[0.62, 0.632]	0.00851	5	0
adaptive	seq2seq	1	0.6254	[0.621, 0.633]	0.00855	5	0
adaptive	seq2seq	10	0.6243	[0.619, 0.631]	0.0079	5	0
adaptive	seq2seq	100	0.6204	[0.602, 0.634]	0.0213	5	0
adaptive	seq2seq	1000	0.6126	[0.591, 0.634]	0.0287	5	0
adaptive	seq2seq	1e+04	0.5901	[0.585, 0.596]	0.0075	5	0
tropical	pointer	0	0.5884	[0.577, 0.599]	0.0144	5	0
tropical	pointer	0.01	0.6102	[0.605, 0.616]	0.00691	5	0
tropical	pointer	0.1	0.5917	[0.571, 0.611]	0.0272	5	0
tropical	pointer	1	0.5973	[0.59, 0.603]	0.0087	5	0
tropical	pointer	10	0.6129	[0.604, 0.624]	0.0128	5	0
tropical	pointer	100	0.5846	[0.55, 0.606]	0.0394	5	0
tropical	pointer	1000	0.6046	[0.592, 0.621]	0.0187	5	0
tropical	pointer	1e+04	0.6025	[0.562, 0.64]	0.0466	5	0
tropical	seq2seq	0	0.6353	[0.598, 0.676]	0.0494	5	0
tropical	seq2seq	0.01	0.6253	[0.615, 0.636]	0.0133	5	0
tropical	seq2seq	0.1	0.6177	[0.592, 0.646]	0.036	5	0
tropical	seq2seq	1	0.6122	[0.604, 0.621]	0.0109	5	0
tropical	seq2seq	10	0.6145	[0.589, 0.635]	0.0308	5	0
tropical	seq2seq	100	0.6184	[0.568, 0.668]	0.0634	5	0
tropical	seq2seq	1000	0.623	[0.596, 0.646]	0.0327	5	0
tropical	seq2seq	1e+04	0.6166	[0.596, 0.636]	0.0264	5	0
vanilla	pointer	0	0.6176	[0.613, 0.623]	0.00585	5	0
vanilla	pointer	0.01	0.6155	[0.61, 0.622]	0.00728	5	0
vanilla	pointer	0.1	0.6159	[0.611, 0.621]	0.00643	5	0
vanilla	pointer	1	0.614	[0.61, 0.618]	0.00505	5	0
vanilla	pointer	10	0.6153	[0.613, 0.618]	0.00276	5	0
vanilla	pointer	100	0.6138	[0.609, 0.617]	0.00597	5	0
vanilla	pointer	1000	0.6105	[0.604, 0.616]	0.00789	5	0
vanilla	pointer	1e+04	0.6131	[0.606, 0.617]	0.00809	5	0
vanilla	seq2seq	0	0.5942	[0.578, 0.613]	0.0226	5	0
vanilla	seq2seq	0.01	0.6007	[0.583, 0.618]	0.0224	5	0
vanilla	seq2seq	0.1	0.6009	[0.578, 0.62]	0.0266	5	0
vanilla	seq2seq	1	0.6034	[0.587, 0.619]	0.0212	5	0
vanilla	seq2seq	10	0.6015	[0.584, 0.617]	0.0216	5	0
vanilla	seq2seq	100	0.601	[0.584, 0.618]	0.023	5	0
vanilla	seq2seq	1000	0.5912	[0.583, 0.606]	0.0159	5	0
vanilla	seq2seq	1e+04	0.5829	[0.582, 0.584]	0.0019	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.002997	[-0.0125, 0.00647]	0.9155
adaptive	pointer	0.1	-0.00236	[-0.0118, 0.00711]	0.9727
adaptive	pointer	1	-0.004524	[-0.014, 0.00494]	0.6446

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	10	-0.003162	[-0.0126, 0.0063]	0.8941	
adaptive	pointer	100	7.766×10^{-4}	[-0.00869, 0.0102]	1.0000	
adaptive	pointer	1000	-0.002202	[-0.0117, 0.00726]	0.9810	
adaptive	pointer	1e+04	9.809×10^{-4}	[-0.00849, 0.0104]	0.9999	
adaptive	seq2seq	0.01	7.714×10^{-4}	[-0.0261, 0.0276]	1.0000	
adaptive	seq2seq	0.1	0.001316	[-0.0255, 0.0282]	1.0000	
adaptive	seq2seq	1	0.002343	[-0.0245, 0.0292]	1.0000	
adaptive	seq2seq	10	0.001266	[-0.0256, 0.0281]	1.0000	
adaptive	seq2seq	100	-0.002697	[-0.0295, 0.0241]	0.9999	
adaptive	seq2seq	1000	-0.01047	[-0.0373, 0.0164]	0.8091	
adaptive	seq2seq	1e+04	-0.03295	[-0.0598, -0.0061]	0.0110	*
tropical	pointer	0.01	0.0218	[-0.0232, 0.0668]	0.6323	
tropical	pointer	0.1	0.003337	[-0.0417, 0.0484]	1.0000	
tropical	pointer	1	0.008903	[-0.0361, 0.0539]	0.9923	
tropical	pointer	10	0.02453	[-0.0205, 0.0696]	0.5152	
tropical	pointer	100	-0.003776	[-0.0488, 0.0413]	1.0000	
tropical	pointer	1000	0.01619	[-0.0288, 0.0612]	0.8582	
tropical	pointer	1e+04	0.01408	[-0.031, 0.0591]	0.9200	
tropical	seq2seq	0.01	-0.01	[-0.0741, 0.0541]	0.9981	
tropical	seq2seq	0.1	-0.01764	[-0.0817, 0.0464]	0.9554	
tropical	seq2seq	1	-0.02306	[-0.0871, 0.041]	0.8579	
tropical	seq2seq	10	-0.02083	[-0.0849, 0.0433]	0.9055	
tropical	seq2seq	100	-0.01694	[-0.081, 0.0471]	0.9633	
tropical	seq2seq	1000	-0.01226	[-0.0763, 0.0518]	0.9936	
tropical	seq2seq	1e+04	-0.0187	[-0.0828, 0.0454]	0.9413	
vanilla	pointer	0.01	-0.002141	[-0.0133, 0.00899]	0.9934	
vanilla	pointer	0.1	-0.00167	[-0.0128, 0.00946]	0.9985	
vanilla	pointer	1	-0.003564	[-0.0147, 0.00757]	0.9114	
vanilla	pointer	10	-0.00227	[-0.0134, 0.00887]	0.9908	
vanilla	pointer	100	-0.003817	[-0.015, 0.00732]	0.8823	
vanilla	pointer	1000	-0.007122	[-0.0183, 0.00401]	0.3516	
vanilla	pointer	1e+04	-0.004503	[-0.0156, 0.00663]	0.7843	
vanilla	seq2seq	0.01	0.006557	[-0.0296, 0.0427]	0.9953	
vanilla	seq2seq	0.1	0.006698	[-0.0294, 0.0428]	0.9947	
vanilla	seq2seq	1	0.00927	[-0.0269, 0.0454]	0.9684	
vanilla	seq2seq	10	0.007349	[-0.0288, 0.0435]	0.9909	
vanilla	seq2seq	100	0.006808	[-0.0293, 0.0429]	0.9941	
vanilla	seq2seq	1000	-0.003001	[-0.0391, 0.0331]	1.0000	
vanilla	seq2seq	1e+04	-0.0113	[-0.0474, 0.0248]	0.9198	

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	6.267×10^{-4}	[3.61×10^{-4}, 0.00108]	0.1624	0.189	0.1564
adaptive	seq2seq	-0.004635	[-0.0058, -0.00347]	0.0015	-0.208	0.1254
tropical	pointer	-3.622×10^{-4}	[-0.00387, 0.00385]	0.8733	-0.0667	0.6331
tropical	seq2seq	-3.291×10^{-4}	[-0.0038, 0.0033]	0.9041	0.0019	1.0000
vanilla	pointer	-6.515×10^{-4}	[-0.00146, 8.11×10^{-5}]	0.2116	-0.0438	0.7415
vanilla	seq2seq	-0.002694	[-0.00571, 3.23×10^{-4}]	0.1085	-0.2	0.1274

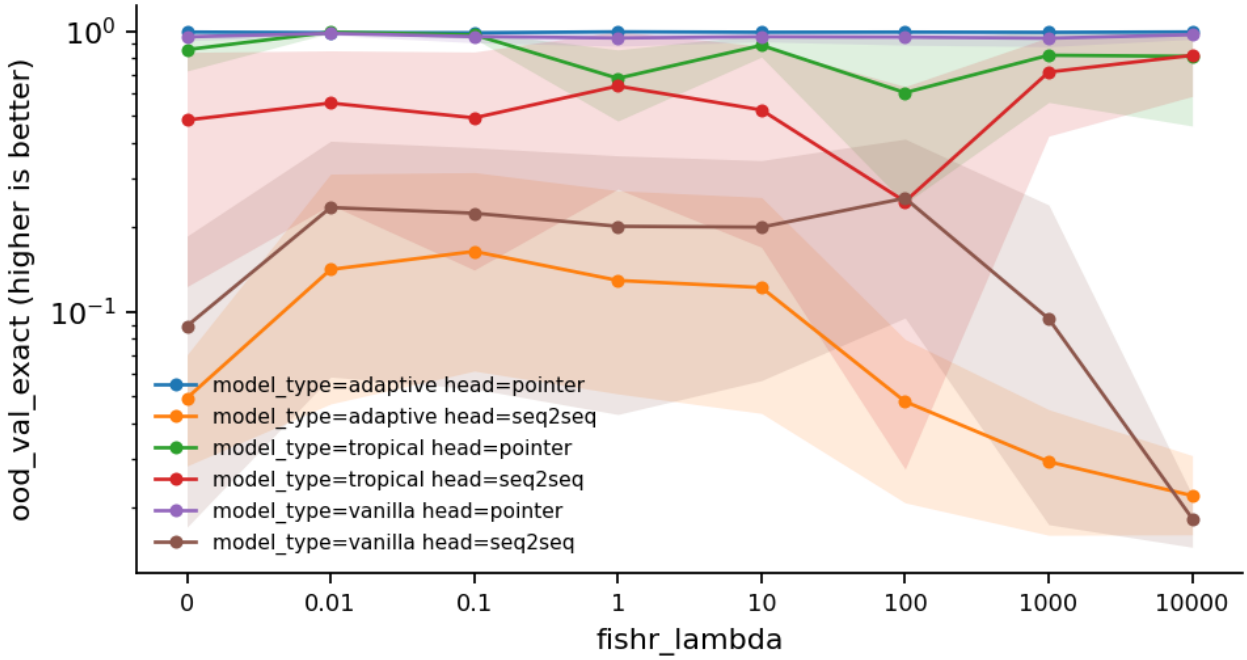


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

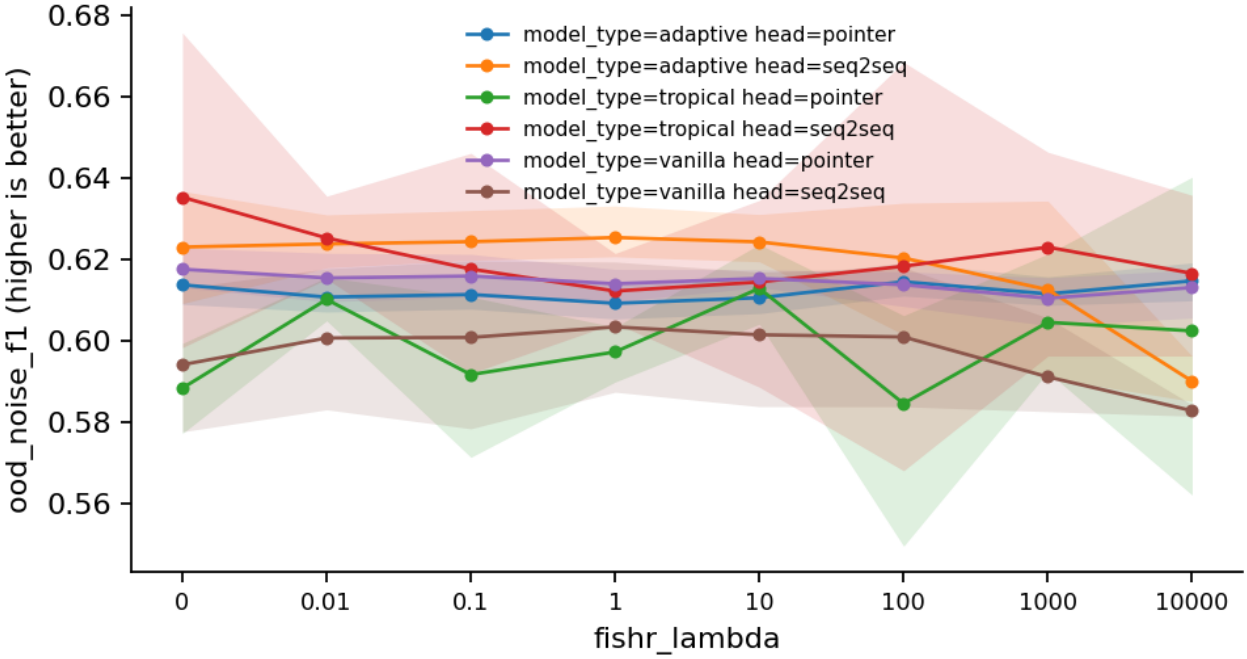


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.02922	[-0.0429, -0.0156]	0.0641
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.005279	[-0.014, 0.00341]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.02421	[-0.0619, 0.0134]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.01675	[-0.0288, -0.00472]	0.2708
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.002425	[-0.0187, 0.0139]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.02918	[-0.0809, 0.0226]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.005907	[-0.0343, 0.0225]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.01064	[-0.0764, 0.0551]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.04113	[-0.0149, 0.0972]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.02458	[-0.00249, 0.0516]	0.7831
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.0168	[-0.0272, 0.0608]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.008809	[-0.0268, 0.0445]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01296	[-0.0342, 0.0601]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.01738	[-0.0427, 0.0775]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.03188	[0.00463, 0.0591]	0.4398
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.03374	$[1.58 \times 10^{-4}, 0.0673]$	0.6414
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.003846	[-0.00642, -0.00127]	0.2002
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.004702	$[-0.00966, 2.53 \times 10^{-4}]$	0.7526
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.004536	[-0.00706, -0.00201]	0.1211
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.004806	$[-0.0103, 6.63 \times 10^{-4}]$	0.8546
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.004739	$[-0.0103, 8.27 \times 10^{-4}]$	0.8546
model_type: adaptive - vanilla head=pointer fishr_lambda=100	7.479×10^{-4}	[-0.00566, 0.00716]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.001074	[-0.00963, 0.0118]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.001637	[-0.00468, 0.00795]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.0289	[0.011, 0.0468]	0.1650
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.02311	[-0.00985, 0.0561]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.02352	[-0.0189, 0.0659]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.02197	[-0.0132, 0.0572]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.02281	[-0.0115, 0.0572]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.01939	[-0.0124, 0.0512]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.02143	[-0.0278, 0.0706]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.007251	[-0.0029, 0.0174]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.009295	[-0.0311, 0.0125]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.01306	[-0.0277, 0.0016]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01297	[-0.0262, 2.89×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01616	[-0.031, -0.00136]	0.8521	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.01372	[-0.0275, 8.65×10^{-5}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.005821	[-0.0386, 0.0269]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.001025	[-0.0411, 0.039]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.02463	[0.0185, 0.0307]	0.0086	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.04691	[-0.108, 0.0138]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.01511	[-0.032, 0.00176]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.02593	[-0.0843, 0.0324]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01495	[-0.037, 0.00713]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.001554	[-0.0326, 0.0295]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.03374	[-0.0909, 0.0234]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.01846	[-0.07, 0.033]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.01413	[-0.0905, 0.0622]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02345	[-0.00158, 0.0485]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01475	[-0.0194, 0.0489]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01508	[-0.022, 0.0522]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01062	[-0.0205, 0.0418]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01383	[-0.0139, 0.0415]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.01282	[-0.0204, 0.046]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01933	[-0.00268, 0.0413]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.03025	[0.0181, 0.0424]	0.0529	

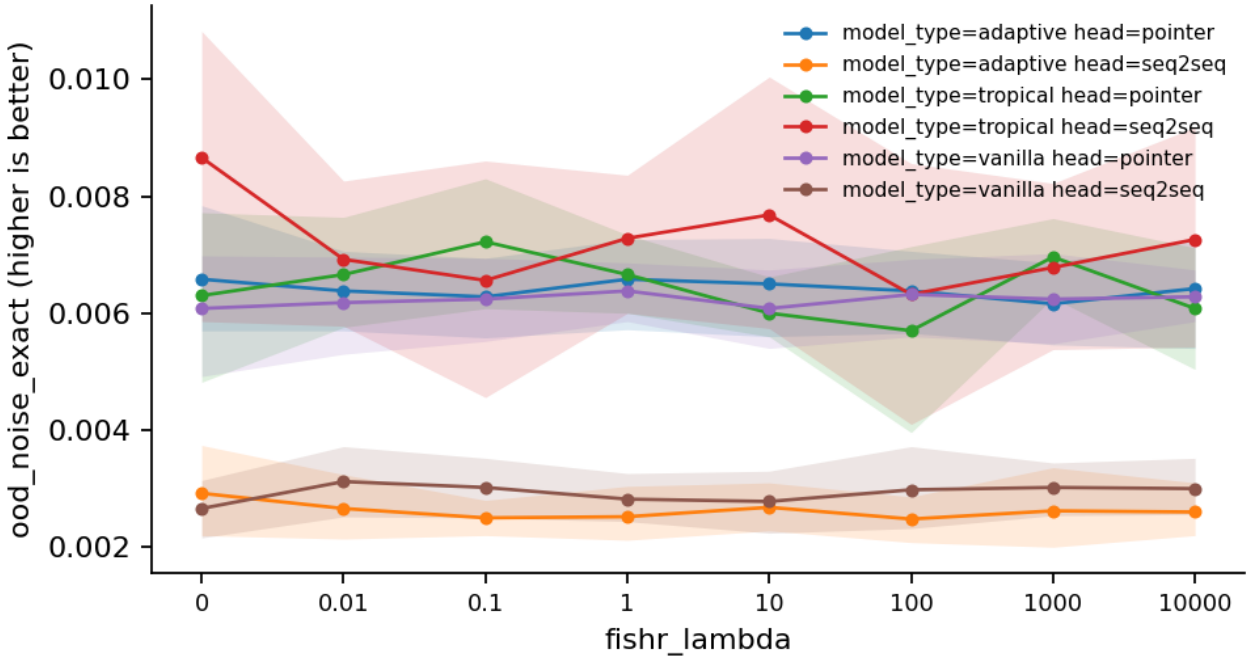


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.00658	[0.0057, 0.00784]	0.00147	5	0
adaptive	pointer	0.01	0.00638	[0.0057, 0.00706]	9.09×10^{-4}	5	0
adaptive	pointer	0.1	0.00628	[0.00558, 0.00694]	8.98×10^{-4}	5	0
adaptive	pointer	1	0.00658	[0.00572, 0.00726]	9.86×10^{-4}	5	0
adaptive	pointer	10	0.0065	[0.0056, 0.00728]	0.00106	5	0
adaptive	pointer	100	0.00638	[0.00566, 0.00706]	9.36×10^{-4}	5	0
adaptive	pointer	1000	0.00616	[0.00546, 0.00686]	9.24×10^{-4}	5	0
adaptive	pointer	1e+04	0.00642	[0.0054, 0.00724]	0.00129	5	0
adaptive	seq2seq	0	0.00292	[0.0022, 0.00374]	9.88×10^{-4}	5	0
adaptive	seq2seq	0.01	0.00266	[0.00214, 0.00324]	7.40×10^{-4}	5	0
adaptive	seq2seq	0.1	0.0025	[0.0022, 0.0028]	3.87×10^{-4}	5	0
adaptive	seq2seq	1	0.00252	[0.00212, 0.00304]	6.06×10^{-4}	5	0
adaptive	seq2seq	10	0.00268	[0.00228, 0.0031]	4.97×10^{-4}	5	0
adaptive	seq2seq	100	0.00248	[0.00208, 0.00286]	5.07×10^{-4}	5	0
adaptive	seq2seq	1000	0.00262	[0.002, 0.00336]	8.76×10^{-4}	5	0
adaptive	seq2seq	1e+04	0.0026	[0.0022, 0.0031]	5.70×10^{-4}	5	0
tropical	pointer	0	0.0063	[0.00482, 0.00772]	0.00202	5	0
tropical	pointer	0.01	0.00666	[0.00576, 0.00764]	0.00122	5	0
tropical	pointer	0.1	0.00722	[0.00608, 0.0083]	0.00146	5	0
tropical	pointer	1	0.00666	[0.006, 0.00734]	8.65×10^{-4}	5	0
tropical	pointer	10	0.006	[0.0056, 0.00662]	6.89×10^{-4}	5	0
tropical	pointer	100	0.0057	[0.00396, 0.00714]	0.002	5	0
tropical	pointer	1000	0.00696	[0.00626, 0.00762]	8.93×10^{-4}	5	0
tropical	pointer	1e+04	0.00608	[0.00504, 0.00712]	0.00133	5	0
tropical	seq2seq	0	0.00866	[0.00586, 0.0108]	0.00326	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.00692	[0.00578, 0.00826]	0.00171	5	0
tropical	seq2seq	0.1	0.00656	[0.00456, 0.0086]	0.00267	5	0
tropical	seq2seq	1	0.00728	[0.006, 0.00836]	0.00152	5	0
tropical	seq2seq	10	0.00768	[0.00574, 0.01]	0.00274	5	0
tropical	seq2seq	100	0.00632	[0.0041, 0.00856]	0.00291	5	0
tropical	seq2seq	1000	0.00678	[0.00538, 0.00822]	0.0019	5	0
tropical	seq2seq	1e+04	0.00726	[0.00542, 0.00918]	0.00236	5	0
vanilla	pointer	0	0.00608	[0.00492, 0.00698]	0.00133	5	0
vanilla	pointer	0.01	0.00618	[0.0053, 0.00696]	0.00105	5	0
vanilla	pointer	0.1	0.00624	[0.00552, 0.00694]	9.69×10^{-4}	5	0
vanilla	pointer	1	0.00638	[0.00586, 0.00686]	6.57×10^{-4}	5	0
vanilla	pointer	10	0.00608	[0.0054, 0.00674]	8.87×10^{-4}	5	0
vanilla	pointer	100	0.00632	[0.0056, 0.00692]	8.35×10^{-4}	5	0
vanilla	pointer	1000	0.00624	[0.00548, 0.00702]	0.001	5	0
vanilla	pointer	1e+04	0.00628	[0.00586, 0.00674]	5.63×10^{-4}	5	0
vanilla	seq2seq	0	0.00266	[0.00216, 0.00314]	6.43×10^{-4}	5	0
vanilla	seq2seq	0.01	0.00312	[0.00252, 0.00372]	8.29×10^{-4}	5	0
vanilla	seq2seq	0.1	0.00302	[0.0025, 0.00352]	6.94×10^{-4}	5	0
vanilla	seq2seq	1	0.00282	[0.00244, 0.00326]	5.12×10^{-4}	5	0
vanilla	seq2seq	10	0.00278	[0.00224, 0.0033]	6.61×10^{-4}	5	0
vanilla	seq2seq	100	0.00298	[0.00232, 0.00372]	9.26×10^{-4}	5	0
vanilla	seq2seq	1000	0.00302	[0.00254, 0.00344]	5.85×10^{-4}	5	0
vanilla	seq2seq	1e+04	0.003	[0.00256, 0.00352]	6.28×10^{-4}	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-1.000×10^{-4}	[-0.00102, 4.20×10^{-4}]	1.0000
adaptive	pointer	0.1	-1.000×10^{-4}	[-0.00128, 4.60×10^{-4}]	1.0000
adaptive	pointer	1	4.000×10^{-4}	[-8.40×10^{-4}, 7.20×10^{-4}]	1.0000
adaptive	pointer	10	3.000×10^{-4}	[-8.60×10^{-4}, 5.40×10^{-4}]	1.0000
adaptive	pointer	100	2.000×10^{-4}	[-9.40×10^{-4}, 3.60×10^{-4}]	1.0000
adaptive	pointer	1000	-1.000×10^{-4}	[-0.00118, 1.80×10^{-4}]	1.0000
adaptive	pointer	1e+04	3.000×10^{-4}	[-9.80×10^{-4}, 6.60×10^{-4}]	1.0000
adaptive	seq2seq	0.01	-2.000×10^{-4}	[-0.00116, 6.40×10^{-4}]	1.0000
adaptive	seq2seq	0.1	-1.000×10^{-4}	[-0.00134, 3.60×10^{-4}]	1.0000
adaptive	seq2seq	1	-2.000×10^{-4}	[-0.00148, 5.60×10^{-4}]	1.0000
adaptive	seq2seq	10	-2.000×10^{-4}	[-0.0011, 5.00×10^{-4}]	1.0000
adaptive	seq2seq	100	1.000×10^{-4}	[-0.00144, 1.80×10^{-4}]	1.0000
adaptive	seq2seq	1000	1.000×10^{-4}	[-0.00146, 5.00×10^{-4}]	1.0000
adaptive	seq2seq	1e+04	1.000×10^{-4}	[-0.00116, 1.40×10^{-4}]	1.0000
tropical	pointer	0.01	5.000×10^{-4}	[-3.00×10^{-4}, 0.001]	1.0000
tropical	pointer	0.1	8.000×10^{-4}	[5.20×10^{-4}, 0.00148]	0.4375
tropical	pointer	1	5.000×10^{-4}	[-6.20×10^{-4}, 0.0014]	1.0000
tropical	pointer	10	-0.001	[-0.00132, 9.80×10^{-4}]	1.0000
tropical	pointer	100	-5.000×10^{-4}	[-0.0018, 5.20×10^{-4}]	1.0000
tropical	pointer	1000	3.000×10^{-4}	[-8.80×10^{-4}, 0.00276]	1.0000
tropical	pointer	1e+04	4.000×10^{-4}	[-0.00196, 8.20×10^{-4}]	1.0000
tropical	seq2seq	0.01	-0.0032	[-0.0048, 0.00132]	1.0000
tropical	seq2seq	0.1	-0.0021	[-0.00482, 8.42×10^{-4}]	1.0000
tropical	seq2seq	1	-0.0011	[-0.00362, 0.00114]	1.0000
tropical	seq2seq	10	-4.000×10^{-4}	[-0.00328, 0.00132]	1.0000

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
tropical	seq2seq	100	-0.0032	[-0.00452, -1.20×10^{-4}]	1.0000
tropical	seq2seq	1000	-0.0022	[-0.00476, 8.00×10^{-4}]	1.0000
tropical	seq2seq	1e+04	-0.0012	[-0.00258, -2.60×10^{-4}]	0.8750
vanilla	pointer	0.01	1.000×10^{-4}	[-0.00106, 0.00142]	1.0000
vanilla	pointer	0.1	-2.000×10^{-4}	[-9.20×10^{-4}, 0.00152]	1.0000
vanilla	pointer	1	0	[-7.60×10^{-4}, 0.00154]	1.0000
vanilla	pointer	10	-1.000×10^{-4}	[-0.00102, 0.00122]	1.0000
vanilla	pointer	100	-1.000×10^{-4}	[-8.20×10^{-4}, 0.00164]	1.0000
vanilla	pointer	1000	-3.000×10^{-4}	[-9.00×10^{-4}, 0.00146]	1.0000
vanilla	pointer	1e+04	0	[-7.60×10^{-4}, 0.00124]	1.0000
vanilla	seq2seq	0.01	3.000×10^{-4}	[2.00×10^{-5}, 9.60×10^{-4}]	1.0000
vanilla	seq2seq	0.1	5.000×10^{-4}	[-2.00×10^{-4}, 8.60×10^{-4}]	1.0000
vanilla	seq2seq	1	1.000×10^{-4}	[-2.20×10^{-4}, 5.20×10^{-4}]	1.0000
vanilla	seq2seq	10	0	[-3.80×10^{-4}, 7.00×10^{-4}]	1.0000
vanilla	seq2seq	100	4.000×10^{-4}	[-3.20×10^{-4}, 8.40×10^{-4}]	1.0000
vanilla	seq2seq	1000	1.000×10^{-4}	[-2.20×10^{-4}, 1.00×10^{-3}]	1.0000
vanilla	seq2seq	1e+04	2.000×10^{-4}	[1.20×10^{-4}, 5.80×10^{-4}]	0.8750

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.05529	[-0.375, 0.185]	0.7524	-0.0514	0.7119
adaptive	seq2seq	-0.03552	[-0.544, 0.523]	0.8395	-0.0343	0.8072
tropical	pointer	-0.1536	[-0.516, 0.0426]	0.3783	-0.122	0.3573
tropical	seq2seq	0.02618	[-0.315, 0.318]	0.8813	0.0133	0.9278
vanilla	pointer	0.002126	[-0.329, 0.327]	0.9903	-0.0019	0.9954
vanilla	seq2seq	-0.01985	[-0.159, 0.129]	0.9099	-0.0171	0.9020

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	2.200×10^{-4}	[-0.00185, 0.00229]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	4.800×10^{-4}	[-0.00112, 0.00208]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	9.800×10^{-4}	[-8.22×10^{-4}, 0.00278]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	2.800×10^{-4}	[-8.66×10^{-4}, 0.00143]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-8.000×10^{-5}	[-0.00155, 0.00139]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-6.200×10^{-4}	[-0.00342, 0.00218]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	7.200×10^{-4}	[-0.00111, 0.00255]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-2.000×10^{-4}	[-0.00123, 8.28×10^{-4}]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.006	[0.00221, 0.00979]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.0038	[0.00125, 0.00635]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.00354	$[1.32 \times 10^{-4}, 0.00695]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.00446	[0.00211, 0.00681]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.0049	$[8.80 \times 10^{-4}, 0.00892]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.00334	$[-8.84 \times 10^{-4}, 0.00756]$	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.00376	[0.0013, 0.00622]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.00426	[0.00156, 0.00696]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	5.000×10^{-4}	[-0.00102, 0.00202]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	2.000×10^{-4}	$[-9.94 \times 10^{-4}, 0.00139]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	4.000×10^{-5}	[-0.00109, 0.00117]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	2.000×10^{-4}	[-0.00115, 0.00155]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	4.200×10^{-4}	$[-9.36 \times 10^{-4}, 0.00178]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	6.000×10^{-5}	[-0.00132, 0.00144]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-8.000×10^{-5}	[-0.00153, 0.00137]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	1.400×10^{-4}	[-0.0013, 0.00158]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	2.600×10^{-4}	[-0.00137, 0.00189]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-4.600×10^{-4}	[-0.00203, 0.00111]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-5.200×10^{-4}	$[-0.00144, 3.98 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-3.000×10^{-4}	$[-0.00134, 7.39 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-1.000×10^{-4}	$[-0.0011, 9.01 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-5.000×10^{-4}	$[-0.00179, 7.93 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-4.000×10^{-4}	$[-0.00116, 3.60 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-4.000×10^{-4}	$[-0.00107, 2.74 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.00366	[0.00153, 0.00579]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.00372	[0.00241, 0.00503]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.00378	[0.00288, 0.00468]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.00406	[0.00279, 0.00533]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.00382	[0.00257, 0.00507]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.0039	[0.0032, 0.0046]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.00354	[0.00286, 0.00422]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.00382	[0.00248, 0.00516]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.00236	[-0.00693, 0.00221]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-2.600×10^{-4}	[-0.0019, 0.00138]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	6.600×10^{-4}	[-0.00191, 0.00323]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-6.200×10^{-4}	[-0.00303, 0.00179]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.00168	[-0.00488, 0.00152]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-6.200×10^{-4}	[-0.00274, 0.0015]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	1.800×10^{-4}	[-0.00204, 0.0024]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.00118	[-0.00353, 0.00117]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.00342	[0.00187, 0.00497]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.00306	[0.00132, 0.0048]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.00322	[0.00212, 0.00432]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.00356	[0.00283, 0.00429]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.0033	[0.0018, 0.0048]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.00334	[0.00172, 0.00496]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.00322	[0.00211, 0.00433]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.00328	[0.00236, 0.0042]	1.0000

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9543	[0.954, 0.955]	0.00105	5	0
adaptive	pointer	0.01	0.9543	[0.954, 0.955]	8.67×10^{-4}	5	0
adaptive	pointer	0.1	0.9544	[0.954, 0.955]	8.61×10^{-4}	5	0
adaptive	pointer	1	0.9544	[0.954, 0.955]	8.85×10^{-4}	5	0
adaptive	pointer	10	0.9544	[0.954, 0.955]	7.79×10^{-4}	5	0
adaptive	pointer	100	0.9544	[0.954, 0.955]	8.35×10^{-4}	5	0
adaptive	pointer	1000	0.9544	[0.954, 0.955]	8.06×10^{-4}	5	0
adaptive	pointer	1e+04	0.9545	[0.954, 0.955]	9.38×10^{-4}	5	0
adaptive	seq2seq	0	0.9464	[0.945, 0.947]	0.00133	5	0
adaptive	seq2seq	0.01	0.9469	[0.946, 0.948]	0.0015	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.9466	[0.946, 0.947]	8.86×10^{-4}	5	0
adaptive	seq2seq	1	0.9467	[0.946, 0.948]	0.00109	5	0
adaptive	seq2seq	10	0.9469	[0.946, 0.948]	0.0014	5	0
adaptive	seq2seq	100	0.9476	[0.945, 0.951]	0.00411	5	0
adaptive	seq2seq	1000	0.9494	[0.946, 0.953]	0.00435	5	0
adaptive	seq2seq	1e+04	0.9538	[0.953, 0.955]	0.00168	5	0
tropical	pointer	0	0.9551	[0.955, 0.956]	7.67×10^{-4}	5	0
tropical	pointer	0.01	0.9552	[0.954, 0.956]	8.82×10^{-4}	5	0
tropical	pointer	0.1	0.9549	[0.954, 0.956]	8.21×10^{-4}	5	0
tropical	pointer	1	0.9552	[0.955, 0.956]	5.96×10^{-4}	5	0
tropical	pointer	10	0.9556	[0.955, 0.956]	5.70×10^{-4}	5	0
tropical	pointer	100	0.955	[0.954, 0.956]	7.12×10^{-4}	5	0
tropical	pointer	1000	0.9549	[0.954, 0.956]	9.27×10^{-4}	5	0
tropical	pointer	1e+04	0.9551	[0.954, 0.956]	0.00102	5	0
tropical	seq2seq	0	0.9506	[0.95, 0.951]	0.00101	5	0
tropical	seq2seq	0.01	0.9516	[0.95, 0.953]	0.00234	5	0
tropical	seq2seq	0.1	0.9514	[0.95, 0.953]	0.00183	5	0
tropical	seq2seq	1	0.9515	[0.95, 0.953]	0.00199	5	0
tropical	seq2seq	10	0.9515	[0.95, 0.953]	0.00161	5	0
tropical	seq2seq	100	0.9515	[0.95, 0.953]	0.00165	5	0
tropical	seq2seq	1000	0.9515	[0.95, 0.953]	0.00144	5	0
tropical	seq2seq	1e+04	0.9513	[0.95, 0.952]	0.00113	5	0
vanilla	pointer	0	0.9543	[0.954, 0.955]	8.38×10^{-4}	5	0
vanilla	pointer	0.01	0.9546	[0.954, 0.955]	7.77×10^{-4}	5	0
vanilla	pointer	0.1	0.9546	[0.954, 0.955]	5.87×10^{-4}	5	0
vanilla	pointer	1	0.9546	[0.954, 0.955]	5.82×10^{-4}	5	0
vanilla	pointer	10	0.9546	[0.954, 0.955]	5.16×10^{-4}	5	0
vanilla	pointer	100	0.9546	[0.954, 0.955]	6.99×10^{-4}	5	0
vanilla	pointer	1000	0.9546	[0.954, 0.955]	6.32×10^{-4}	5	0
vanilla	pointer	1e+04	0.9547	[0.954, 0.955]	6.05×10^{-4}	5	0
vanilla	seq2seq	0	0.9507	[0.946, 0.955]	0.00551	5	0
vanilla	seq2seq	0.01	0.9492	[0.946, 0.953]	0.00453	5	0
vanilla	seq2seq	0.1	0.9492	[0.946, 0.953]	0.00444	5	0
vanilla	seq2seq	1	0.9494	[0.946, 0.953]	0.00462	5	0
vanilla	seq2seq	10	0.9493	[0.946, 0.953]	0.00441	5	0
vanilla	seq2seq	100	0.9494	[0.946, 0.953]	0.00451	5	0
vanilla	seq2seq	1000	0.9526	[0.949, 0.955]	0.00403	5	0
vanilla	seq2seq	1e+04	0.9545	[0.954, 0.955]	9.10×10^{-4}	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	4.184×10^{-5}	[-0.0015, 0.00158]	1.0000
adaptive	pointer	0.1	7.518×10^{-5}	[-0.00146, 0.00161]	1.0000
adaptive	pointer	1	6.751×10^{-5}	[-0.00147, 0.00161]	1.0000
adaptive	pointer	10	1.227×10^{-4}	[-0.00142, 0.00166]	1.0000
adaptive	pointer	100	8.134×10^{-5}	[-0.00146, 0.00162]	1.0000
adaptive	pointer	1000	9.238×10^{-5}	[-0.00145, 0.00163]	1.0000
adaptive	pointer	1e+04	2.109×10^{-4}	[-0.00133, 0.00175]	0.9992
adaptive	seq2seq	0.01	5.085×10^{-4}	[-0.00371, 0.00473]	0.9996
adaptive	seq2seq	0.1	2.769×10^{-4}	[-0.00394, 0.0045]	1.0000
adaptive	seq2seq	1	3.747×10^{-4}	[-0.00384, 0.00459]	1.0000
adaptive	seq2seq	10	5.074×10^{-4}	[-0.00371, 0.00473]	0.9996

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	100	0.001233	[-0.00299, 0.00545]	0.9409	
adaptive	seq2seq	1000	0.003039	[-0.00118, 0.00726]	0.2417	
adaptive	seq2seq	1e+04	0.007408	[0.00319, 0.0116]	0.0002	*
tropical	pointer	0.01	5.087×10^{-5}	[-0.00135, 0.00145]	1.0000	
tropical	pointer	0.1	-1.889×10^{-4}	[-0.00159, 0.00121]	0.9992	
tropical	pointer	1	4.231×10^{-5}	[-0.00136, 0.00144]	1.0000	
tropical	pointer	10	4.716×10^{-4}	$[-9.27 \times 10^{-4}$, 0.00187]	0.8901	
tropical	pointer	100	-9.867×10^{-5}	[-0.0015, 0.0013]	1.0000	
tropical	pointer	1000	-1.850×10^{-4}	[-0.00158, 0.00121]	0.9993	
tropical	pointer	1e+04	-5.287×10^{-5}	[-0.00145, 0.00135]	1.0000	
tropical	seq2seq	0.01	0.001073	[-0.00185, 0.004]	0.8473	
tropical	seq2seq	0.1	7.945×10^{-4}	[-0.00213, 0.00372]	0.9583	
tropical	seq2seq	1	9.333×10^{-4}	[-0.00199, 0.00386]	0.9129	
tropical	seq2seq	10	9.241×10^{-4}	[-0.002, 0.00385]	0.9165	
tropical	seq2seq	100	9.379×10^{-4}	[-0.00199, 0.00387]	0.9110	
tropical	seq2seq	1000	9.176×10^{-4}	[-0.00201, 0.00384]	0.9190	
tropical	seq2seq	1e+04	7.480×10^{-4}	[-0.00218, 0.00368]	0.9691	
vanilla	pointer	0.01	3.256×10^{-4}	$[-8.32 \times 10^{-4}$, 0.00148]	0.9504	
vanilla	pointer	0.1	3.339×10^{-4}	$[-8.23 \times 10^{-4}$, 0.00149]	0.9443	
vanilla	pointer	1	3.962×10^{-4}	$[-7.61 \times 10^{-4}$, 0.00155]	0.8830	
vanilla	pointer	10	3.947×10^{-4}	$[-7.63 \times 10^{-4}$, 0.00155]	0.8847	
vanilla	pointer	100	3.166×10^{-4}	$[-8.41 \times 10^{-4}$, 0.00147]	0.9567	
vanilla	pointer	1000	3.953×10^{-4}	$[-7.62 \times 10^{-4}$, 0.00155]	0.8840	
vanilla	pointer	1e+04	4.840×10^{-4}	$[-6.73 \times 10^{-4}$, 0.00164]	0.7593	
vanilla	seq2seq	0.01	-0.001518	[-0.00905, 0.00601]	0.9914	
vanilla	seq2seq	0.1	-0.001464	[-0.009, 0.00607]	0.9930	
vanilla	seq2seq	1	-0.001303	[-0.00883, 0.00623]	0.9965	
vanilla	seq2seq	10	-0.001396	[-0.00893, 0.00614]	0.9947	
vanilla	seq2seq	100	-0.001317	[-0.00885, 0.00621]	0.9962	
vanilla	seq2seq	1000	0.001896	[-0.00564, 0.00943]	0.9713	
vanilla	seq2seq	1e+04	0.003806	[-0.00373, 0.0113]	0.5909	

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	1.983×10^{-5}	$[-1.87 \times 10^{-5}, 5.65 \times 10^{-5}]$	0.7677	0.0514	0.7015
adaptive	seq2seq	9.671×10^{-4}	$[7.60 \times 10^{-4}$, 0.00131]	0.0002	0.284	0.0358
tropical	pointer	-1.588×10^{-5}	$[-8.35 \times 10^{-5}, 2.28 \times 10^{-5}]$	0.8093	-0.0514	0.7101
tropical	seq2seq	-2.590×10^{-5}	$[-1.85 \times 10^{-4}, 1.38 \times 10^{-4}]$	0.8507	-0.0514	0.7249
vanilla	pointer	1.850×10^{-5}	$[5.44 \times 10^{-7}, 3.89 \times 10^{-5}]$	0.7106	0.0438	0.7560
vanilla	seq2seq	8.099×10^{-4}	$[2.67 \times 10^{-4}$, 0.00135]	0.0215	0.28	0.0352

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	8.558×10^{-4}	$[1.63 \times 10^{-4}$, 0.00155]	0.3458
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	5.810×10^{-4}	$[1.68 \times 10^{-4}, 9.94 \times 10^{-4}]$	0.2451

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	3.331×10^{-4}	$[-3.50 \times 10^{-5}, 7.01 \times 10^{-4}]$	0.6588	
model_type: tropical - vanilla head=pointer fishr_lambda=1	5.019×10^{-4}	$[8.16 \times 10^{-5}, 9.22 \times 10^{-4}]$	0.3540	
model_type: tropical - vanilla head=pointer fishr_lambda=10	9.327×10^{-4}	$[6.51 \times 10^{-4}, 0.00121]$	0.0124	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	4.406×10^{-4}	$[4.59 \times 10^{-5}, 8.35 \times 10^{-4}]$	0.3986	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	2.754×10^{-4}	$[-2.81 \times 10^{-4}, 8.32 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	3.190×10^{-4}	$[-3.18 \times 10^{-4}, 9.56 \times 10^{-4}]$	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-1.241×10^{-4}	[-0.0062, 0.00595]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.002467	[-0.00449, 0.00942]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.002134	[-0.00404, 0.00831]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.002112	[-0.00454, 0.00876]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.002196	[-0.0039, 0.0083]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.002131	[-0.00382, 0.00808]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.001103	[-0.00727, 0.00507]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.003182	[-0.00433, -0.00203]	0.0233	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	4.603×10^{-5}	$[-8.79 \times 10^{-4}, 9.71 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-2.378×10^{-4}	$[-5.31 \times 10^{-4}, 5.54 \times 10^{-5}]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-2.127×10^{-4}	$[-6.03 \times 10^{-4}, 1.77 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-2.826×10^{-4}	$[-6.72 \times 10^{-4}, 1.07 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-2.260×10^{-4}	$[-5.64 \times 10^{-4}, 1.12 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-1.892×10^{-4}	$[-4.82 \times 10^{-4}, 1.04 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-2.569×10^{-4}	$[-4.91 \times 10^{-4}, -2.27 \times 10^{-5}]$	0.6114	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-2.271×10^{-4}	$[-9.04 \times 10^{-4}, 4.50 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.004315	[-0.0102, 0.00153]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.002288	[-0.00778, 0.00321]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.002574	[-0.0084, 0.00325]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.002637	[-0.00857, 0.00329]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.002411	[-0.00766, 0.00283]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.001765	[-0.00609, 0.00256]	1.0000	

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.003172	[-0.0114, 0.00504]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-7.130×10^{-4}	[-0.00183, 4.02×10^{-4}]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.00793	[0.00621, 0.00965]	0.0038	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.007464	[0.00614, 0.00879]	0.0019	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.007729	[0.0066, 0.00886]	0.0010	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.007623	[0.00672, 0.00852]	0.0005	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.007546	[0.00625, 0.00884]	0.0018	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.006779	[0.00208, 0.0115]	0.1769	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.004984	[3.82×10^{-4}, 0.00959]	0.3968	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	7.336×10^{-4}	[-4.80×10^{-4}, 0.00195]	0.6744	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.00455	[0.00387, 0.00523]	0.0011	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.003527	[0.00138, 0.00568]	0.1245	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.003566	[0.00194, 0.00519]	0.0513	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.003659	[0.00175, 0.00557]	0.0782	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.004097	[0.00277, 0.00542]	0.0172	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.003513	[0.00225, 0.00478]	0.0244	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.003447	[0.0021, 0.00479]	0.0310	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.003749	[0.00307, 0.00443]	0.0020	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.00357	[-0.00286, 0.00999]	0.6744	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.005414	[-3.68×10^{-4}, 0.0112]	0.4815	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.005367	[-1.30×10^{-4}, 0.0109]	0.4815	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.005269	[-4.51×10^{-4}, 0.011]	0.4815	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.005361	[-1.32×10^{-4}, 0.0109]	0.4815	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.005203	[-5.34×10^{-4}, 0.0109]	0.4815	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.002069	[-0.0032, 0.00734]	0.6744	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	2.476×10^{-4}	[-8.22×10^{-4}, 0.00132]	0.6744	

A.7 `id_exact`

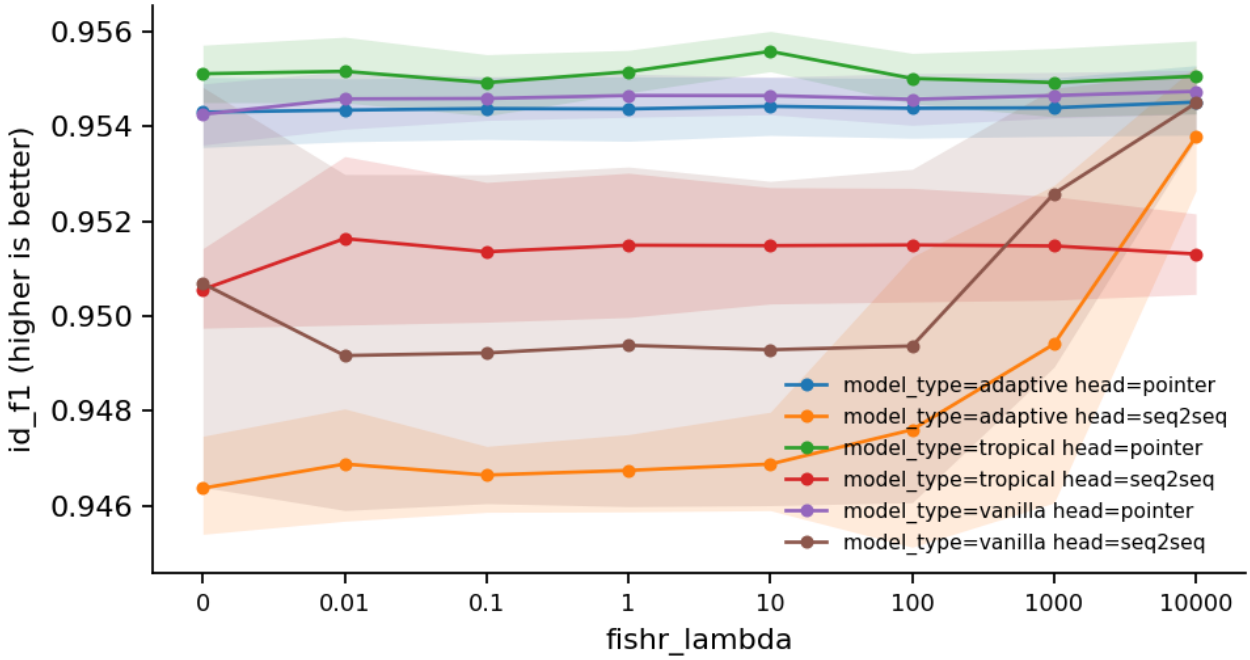


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

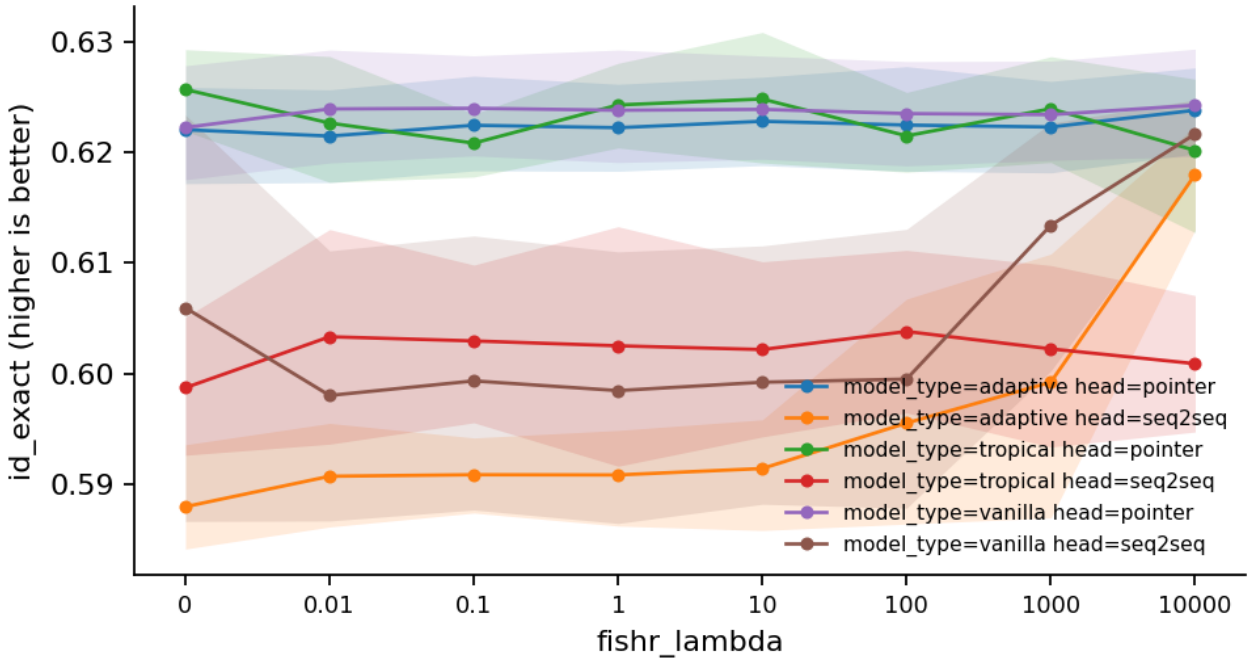


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6221	[0.617, 0.626]	0.00574	5	0
adaptive	pointer	0.01	0.6215	[0.617, 0.626]	0.00554	5	0
adaptive	pointer	0.1	0.6225	[0.618, 0.627]	0.00546	5	0
adaptive	pointer	1	0.6223	[0.618, 0.626]	0.00522	5	0
adaptive	pointer	10	0.6228	[0.619, 0.627]	0.00531	5	0
adaptive	pointer	100	0.6225	[0.618, 0.628]	0.00615	5	0
adaptive	pointer	1000	0.6223	[0.618, 0.626]	0.00534	5	0
adaptive	pointer	1e+04	0.6238	[0.62, 0.628]	0.00497	5	0
adaptive	seq2seq	0	0.5879	[0.584, 0.594]	0.0064	5	0
adaptive	seq2seq	0.01	0.5907	[0.586, 0.595]	0.00606	5	0
adaptive	seq2seq	0.1	0.5908	[0.587, 0.594]	0.00474	5	0
adaptive	seq2seq	1	0.5908	[0.586, 0.595]	0.00578	5	0
adaptive	seq2seq	10	0.5914	[0.586, 0.596]	0.00664	5	0
adaptive	seq2seq	100	0.5955	[0.586, 0.607]	0.0131	5	0
adaptive	seq2seq	1000	0.5992	[0.587, 0.611]	0.0156	5	0
adaptive	seq2seq	1e+04	0.618	[0.613, 0.622]	0.00631	5	0
tropical	pointer	0	0.6257	[0.622, 0.629]	0.00488	5	0
tropical	pointer	0.01	0.6227	[0.617, 0.629]	0.00744	5	0
tropical	pointer	0.1	0.6209	[0.618, 0.624]	0.00372	5	0
tropical	pointer	1	0.6243	[0.62, 0.628]	0.00509	5	0
tropical	pointer	10	0.6249	[0.619, 0.631]	0.00774	5	0
tropical	pointer	100	0.6215	[0.618, 0.625]	0.00461	5	0
tropical	pointer	1000	0.624	[0.619, 0.629]	0.0064	5	0
tropical	pointer	1e+04	0.6202	[0.613, 0.627]	0.00932	5	0
tropical	seq2seq	0	0.5987	[0.593, 0.605]	0.00808	5	0
tropical	seq2seq	0.01	0.6033	[0.594, 0.613]	0.013	5	0
tropical	seq2seq	0.1	0.6029	[0.596, 0.61]	0.00935	5	0
tropical	seq2seq	1	0.6025	[0.592, 0.613]	0.0143	5	0
tropical	seq2seq	10	0.6022	[0.594, 0.61]	0.0102	5	0
tropical	seq2seq	100	0.6038	[0.596, 0.611]	0.0097	5	0
tropical	seq2seq	1000	0.6022	[0.593, 0.61]	0.00999	5	0
tropical	seq2seq	1e+04	0.6009	[0.595, 0.607]	0.00787	5	0
vanilla	pointer	0	0.6223	[0.618, 0.628]	0.00657	5	0
vanilla	pointer	0.01	0.624	[0.619, 0.629]	0.00656	5	0
vanilla	pointer	0.1	0.624	[0.62, 0.629]	0.00585	5	0
vanilla	pointer	1	0.6238	[0.619, 0.629]	0.00643	5	0
vanilla	pointer	10	0.6239	[0.619, 0.629]	0.00599	5	0
vanilla	pointer	100	0.6235	[0.619, 0.628]	0.00629	5	0
vanilla	pointer	1000	0.6234	[0.619, 0.628]	0.00573	5	0
vanilla	pointer	1e+04	0.6243	[0.62, 0.629]	0.00611	5	0
vanilla	seq2seq	0	0.6059	[0.587, 0.623]	0.0239	5	0
vanilla	seq2seq	0.01	0.598	[0.587, 0.611]	0.0156	5	0
vanilla	seq2seq	0.1	0.5993	[0.588, 0.612]	0.016	5	0
vanilla	seq2seq	1	0.5984	[0.586, 0.611]	0.0161	5	0
vanilla	seq2seq	10	0.5992	[0.588, 0.612]	0.0152	5	0
vanilla	seq2seq	100	0.5995	[0.588, 0.613]	0.016	5	0
vanilla	seq2seq	1000	0.6134	[0.6, 0.623]	0.0149	5	0
vanilla	seq2seq	1e+04	0.6217	[0.618, 0.625]	0.00465	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-5.800×10^{-4}	[-0.0101, 0.00899]	1.0000	
adaptive	pointer	0.1	4.000×10^{-4}	[-0.00917, 0.00997]	1.0000	
adaptive	pointer	1	1.800×10^{-4}	[-0.00939, 0.00975]	1.0000	
adaptive	pointer	10	7.600×10^{-4}	[-0.00881, 0.0103]	1.0000	
adaptive	pointer	100	4.200×10^{-4}	[-0.00915, 0.00999]	1.0000	
adaptive	pointer	1000	2.400×10^{-4}	[-0.00933, 0.00981]	1.0000	
adaptive	pointer	1e+04	0.00176	[-0.00781, 0.0113]	0.9949	
adaptive	seq2seq	0.01	0.00276	[-0.0128, 0.0183]	0.9958	
adaptive	seq2seq	0.1	0.0029	[-0.0126, 0.0184]	0.9944	
adaptive	seq2seq	1	0.00288	[-0.0126, 0.0184]	0.9946	
adaptive	seq2seq	10	0.00346	[-0.0121, 0.019]	0.9849	
adaptive	seq2seq	100	0.00758	[-0.00795, 0.0231]	0.6243	
adaptive	seq2seq	1000	0.01128	[-0.00425, 0.0268]	0.2346	
adaptive	seq2seq	1e+04	0.03008	[0.0146, 0.0456]	$< 10^{-4}$	*
tropical	pointer	0.01	-0.00304	[-0.0142, 0.00814]	0.9580	
tropical	pointer	0.1	-0.00486	[-0.016, 0.00632]	0.7286	
tropical	pointer	1	-0.0014	[-0.0126, 0.00978]	0.9995	
tropical	pointer	10	-8.600×10^{-4}	[-0.012, 0.0103]	1.0000	
tropical	pointer	100	-0.0042	[-0.0154, 0.00698]	0.8332	
tropical	pointer	1000	-0.00174	[-0.0129, 0.00944]	0.9981	
tropical	pointer	1e+04	-0.00552	[-0.0167, 0.00566]	0.6138	
tropical	seq2seq	0.01	0.00462	[-0.0137, 0.023]	0.9714	
tropical	seq2seq	0.1	0.00424	[-0.0141, 0.0226]	0.9817	
tropical	seq2seq	1	0.0038	[-0.0146, 0.0222]	0.9900	
tropical	seq2seq	10	0.00346	[-0.0149, 0.0218]	0.9941	
tropical	seq2seq	100	0.0051	[-0.0133, 0.0235]	0.9534	
tropical	seq2seq	1000	0.00352	[-0.0148, 0.0219]	0.9935	
tropical	seq2seq	1e+04	0.00218	[-0.0162, 0.0205]	0.9997	
vanilla	pointer	0.01	0.00168	[-0.00915, 0.0125]	0.9982	
vanilla	pointer	0.1	0.00174	[-0.00909, 0.0126]	0.9977	
vanilla	pointer	1	0.00156	[-0.00927, 0.0124]	0.9988	
vanilla	pointer	10	0.00164	[-0.00919, 0.0125]	0.9984	
vanilla	pointer	100	0.00126	[-0.00957, 0.0121]	0.9997	
vanilla	pointer	1000	0.00116	[-0.00967, 0.012]	0.9998	
vanilla	pointer	1e+04	0.00202	[-0.00881, 0.0128]	0.9945	
vanilla	seq2seq	0.01	-0.00794	[-0.036, 0.0201]	0.9487	
vanilla	seq2seq	0.1	-0.00662	[-0.0346, 0.0214]	0.9793	
vanilla	seq2seq	1	-0.0075	[-0.0355, 0.0205]	0.9610	
vanilla	seq2seq	10	-0.00674	[-0.0348, 0.0213]	0.9773	
vanilla	seq2seq	100	-0.00646	[-0.0345, 0.0216]	0.9818	
vanilla	seq2seq	1000	0.00748	[-0.0205, 0.0355]	0.9615	
vanilla	seq2seq	1e+04	0.01574	[-0.0123, 0.0438]	0.4832	

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	2.479×10^{-4}	$[1.25 \times 10^{-4}, 3.71 \times 10^{-4}]$	0.5637	0.112	0.3985
adaptive	seq2seq	0.003694	[0.00282, 0.0052]	0.0001	0.389	0.0032
tropical	pointer	-1.429×10^{-4}	$[-7.09 \times 10^{-4}, 2.54 \times 10^{-4}]$	0.7904	0.00762	0.9650
tropical	seq2seq	-2.664×10^{-4}	$[-8.45 \times 10^{-4}, 3.24 \times 10^{-4}]$	0.7543	-0.0971	0.4731
vanilla	pointer	-1.571×10^{-5}	$[-8.86 \times 10^{-5}, 4.43 \times 10^{-5}]$	0.9740	0.0114	0.9416

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	0.003581	[0.00108, 0.00609]	0.0058	0.427	0.0018

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.00344	[-0.00118, 0.00806]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.00128	[-0.0039, 0.00134]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.00316	[-0.0083, 0.00198]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	4.800×10^{-4}	[-0.00205, 0.00301]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	9.400×10^{-4}	[-0.00295, 0.00483]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.00202	[-0.00484, 8.04×10^{-4}]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	5.400×10^{-4}	[-0.00122, 0.0023]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.0041	[-0.0126, 0.0044]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.00724	[-0.0376, 0.0231]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.00532	[-0.0225, 0.0331]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.00362	[-0.0216, 0.0289]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.00406	[-0.0262, 0.0343]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.00296	[-0.0227, 0.0287]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.00432	[-0.0203, 0.029]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0112	[-0.0376, 0.0152]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.0208	[-0.0309, -0.0107]	0.0750
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-2.000×10^{-4}	[-0.00598, 0.00558]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.00246	[-0.00393, -9.89×10^{-4}]	0.1553
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.00154	[-0.0025, -5.80×10^{-4}]	0.1683
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.00158	[-0.00349, 3.34×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.00108	[-0.00278, 6.21×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.00104	[-0.00261, 5.32×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.00112	[-0.00244, 2.04×10^{-4}]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-4.600×10^{-4}	[-0.00241, 0.00149]	1.0000

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.01802	[-0.0454, 0.00935]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00732	[-0.0315, 0.0168]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.0085	[-0.0333, 0.0163]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00764	[-0.0329, 0.0176]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.00782	[-0.0314, 0.0158]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00398	[-0.022, 0.0141]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.01422	[-0.0466, 0.0182]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00368	[-0.0101, 0.00273]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.03416	[0.0226, 0.0457]	0.0214	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.03082	[0.0271, 0.0346]	0.0005	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.03166	[0.0265, 0.0368]	0.0016	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.03146	[0.0286, 0.0343]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.03146	[0.0254, 0.0376]	0.0028	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.027	[0.0101, 0.0439]	0.1486	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02312	[0.00346, 0.0428]	0.3093	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.00584	$[-6.78 \times 10^{-4}, 0.0124]$	0.4227	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.02702	[0.02, 0.034]	0.0082	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01936	[0.00605, 0.0327]	0.1719	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.01792	[0.00777, 0.0281]	0.1205	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.02182	[0.00874, 0.0349]	0.1371	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.0227	[0.0184, 0.027]	0.0026	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01772	[0.00829, 0.0272]	0.1032	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.02176	[0.0139, 0.0297]	0.0268	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.01932	[0.00656, 0.0321]	0.1640	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01634	[-0.0123, 0.0449]	0.5636	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.02596	$[5.62 \times 10^{-4}, 0.0514]$	0.4227	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0247	$[-5.03 \times 10^{-4}, 0.0499]$	0.4227	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.0254	$[-4.21 \times 10^{-5}, 0.0508]$	0.4227	

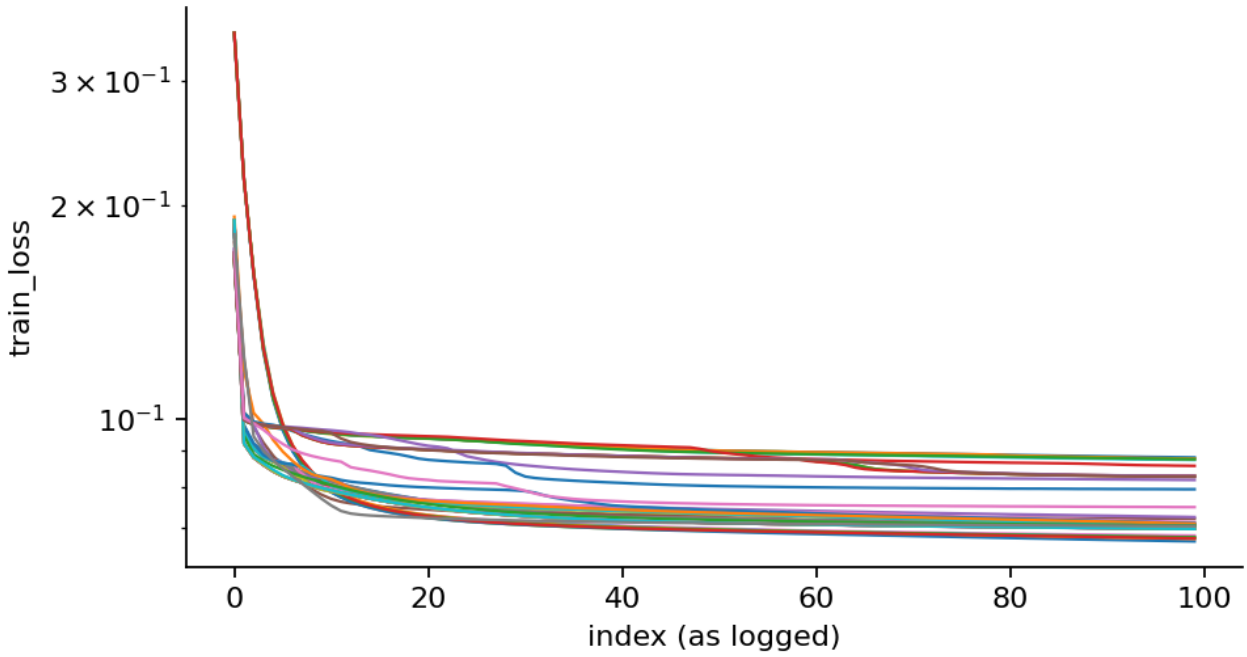


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.02472	$[1.19 \times 10^{-5}$, 0.0494]	0.4227
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02406	[-0.00143, 0.0495]	0.4227
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01002	[-0.0111, 0.0312]	0.5636
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.00262	[-0.00402, 0.00926]	0.5636

B Curves

C Configuration

The base configuration every run shares (frozen hash `765d817c3d73`); factor levels are in Table 1.

```
task:
  name: FloydWarshallDataset
  value_range:
  - 1
  - 15
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 20
  noise_prob: 0.0
data:
  train_lengths:
```

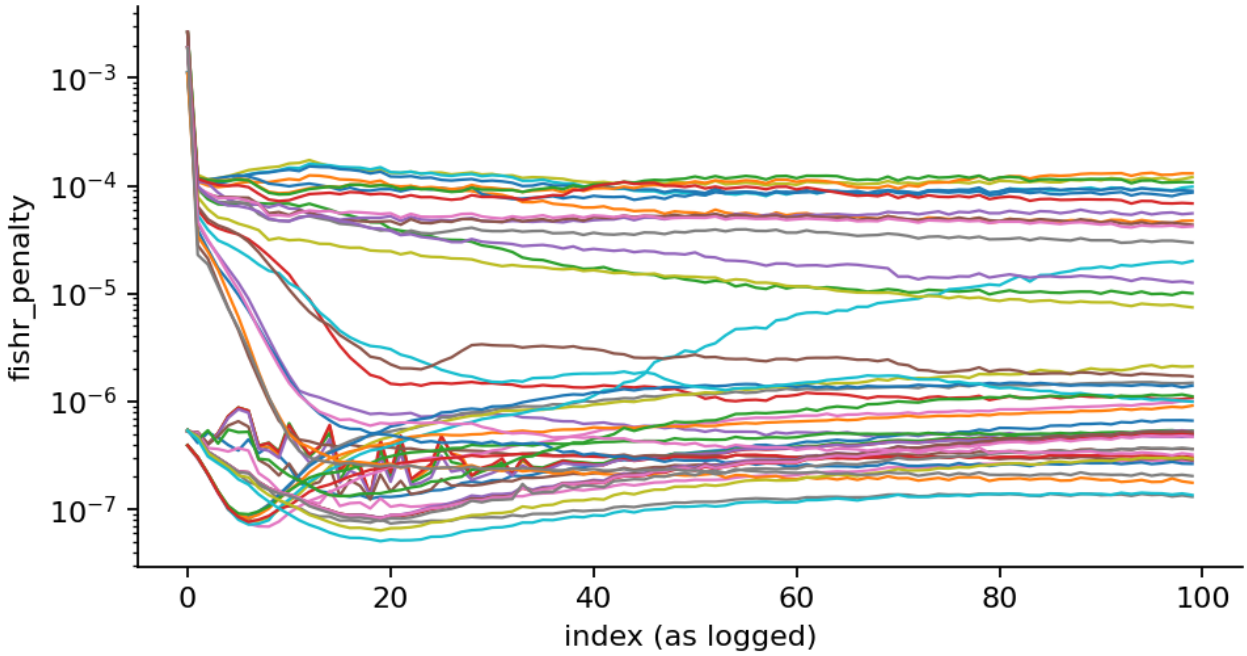


Figure 10: `fishr_penalty`, the mean over seeds per cell.

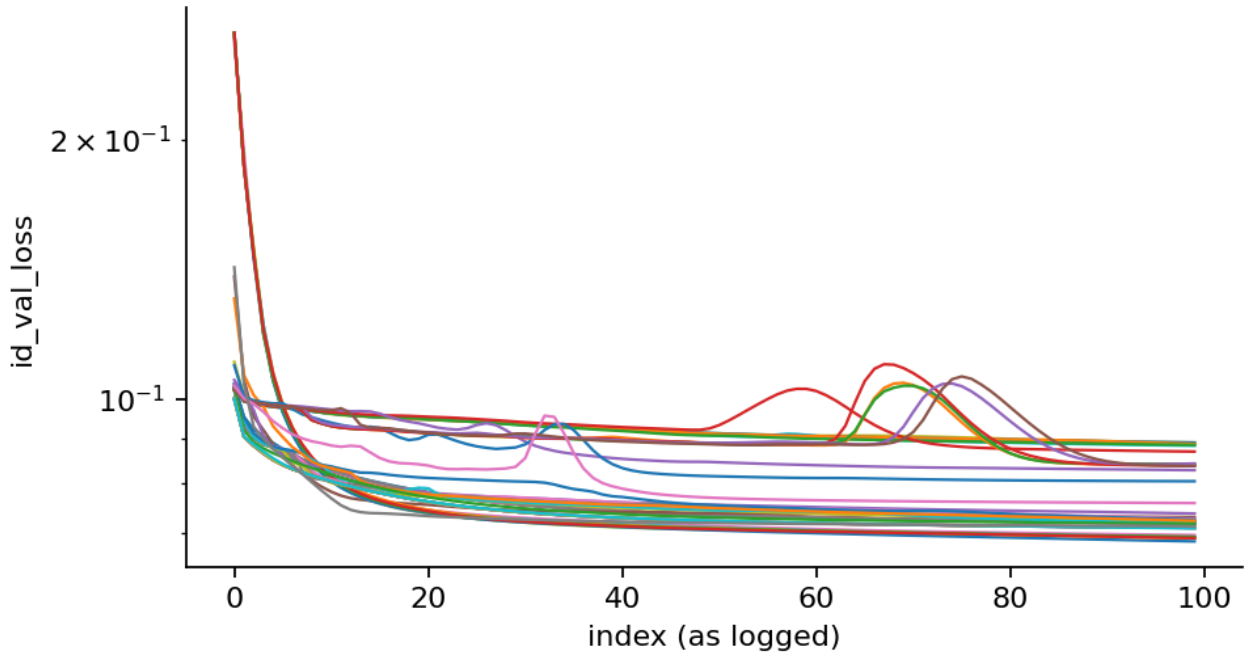


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  ood_value_range:
  - 16
  - 30
  ood_noise_range:
  - 1
  - 10
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 8
  eval_batch_size: 8
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/FloydWarshall/results`.